Three-Bond Extension

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1 Model and conventions

This note fixes one real-space convention and derives the corresponding flux-free Majorana mean-field theory. The order of the derivation is deliberately concrete: first the spin Hamiltonian is written in terms of bond operators, then the unit-cell displacements are fixed, and only then are the Fourier phases and SCF equations derived. The diagnostic chiral NNN hopping κ is not part of the self-consistency problem; it is inserted only after the $\kappa = 0$ fixed point has been obtained.

1.1 Hamiltonian

The microscopic model considered in this note is

$$H = H_K + H_J, \tag{1}$$

with the nearest-neighbor Kitaev part

$$H_K = K_1 \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} \sigma_A^\alpha(\mathbf{r})\, \sigma_B^\alpha(\mathbf{r} + \boldsymbol{\Delta}_\alpha), \tag{2}$$

and the bond-pair four-spin interaction

$$H_J = J \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} \left[\sigma_A^\alpha(\mathbf{r})\, \sigma_B^\alpha(\mathbf{r} + \boldsymbol{\Delta}_\alpha)\right] \left[\sigma_A^\alpha(\mathbf{r} + \mathbf{p}_\alpha)\, \sigma_B^\alpha(\mathbf{r} + \mathbf{p}_\alpha + \boldsymbol{\Delta}_\alpha)\right]. \tag{3}$$

Here $\mathbf{r}$ labels the A-sublattice unit cell. The vector $\boldsymbol{\Delta}_\alpha$ is the A-to-B cell displacement of an α-bond, and $\mathbf{p}_\alpha$ is the Bravais displacement between the two parallel α-bonds forming the bond pair. These vectors are defined geometrically in Subsection 1.2; Eqs. (2) and (3) are written first so that the object being mean-field-decoupled is fixed before introducing the lattice bookkeeping.

For the Kitaev Majorana representation, each spin is written in terms of four Majorana operators,

$$\sigma_j^\alpha = i b_j^\alpha c_j. \tag{4}$$

For an oriented bond (j, k), define

$$u_{jk}^\alpha = i b_j^\alpha b_k^\alpha = -u_{kj}^\alpha. \tag{5}$$

We orient each Kitaev bond from the A sublattice to the B sublattice and choose the uniform flux-free gauge

$$u_{BA}^\alpha = +1, \qquad u_{AB}^\alpha = -1. \tag{6}$$

With this convention, an A-to-B α-bond satisfies

$$\sigma_A^\alpha(\mathbf{r})\, \sigma_B^\alpha(\mathbf{r} + \boldsymbol{\Delta}_\alpha) \longrightarrow i c_A(\mathbf{r}) c_B(\mathbf{r} + \boldsymbol{\Delta}_\alpha). \tag{7}$$

Consequently the Kitaev term becomes

$$H_K \longrightarrow K_1 \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} i c_A(\mathbf{r}) c_B(\mathbf{r} + \boldsymbol{\Delta}_\alpha), \tag{8}$$

and the four-spin interaction becomes

$$H_J \longrightarrow J \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} \Big[ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\Big]\Big[ic_A(\mathbf{r}+\mathbf{p}_\alpha)c_B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha)\Big]. \tag{9}$$

The sign in (9) can be checked directly. Each spin bond gives ic_Ac_B, so the product of two bond operators is already

$$\big[ic_Ac_B\big]\big[ic_Ac_B\big]. \tag{10}$$

Thus no additional sign is hidden in passing from (3) to (9). From this point onward the calculation is performed in the flux-free Majorana form (8)–(9); hats on operators and the fixed gauge variables are omitted.

The diagnostic κ term is not included in (1). It is a post-SCF single-particle chiral NNN Majorana hopping used to gently gap the $\kappa = 0$ spectrum for Chern-number diagnostics. Its real-space orientation and Fourier form are fixed separately in Subsection 2.2.

1.2 Geometry

Set the NN distance to unity. The three NN vectors are

$$\mathbf{s}_z = (0,-1), \quad \mathbf{s}_x = \Big(\tfrac{\sqrt{3}}{2}, \tfrac{1}{2}\Big), \quad \mathbf{s}_y = \Big(-\tfrac{\sqrt{3}}{2}, \tfrac{1}{2}\Big), \tag{11}$$

and the Bravais vectors are

$$\mathbf{R}_1 = (\sqrt{3}, 0), \qquad \mathbf{R}_2 = \Big(\tfrac{\sqrt{3}}{2}, \tfrac{3}{2}\Big). \tag{12}$$

Each unit cell holds one A site and one B site, indexed by $(n_1, \ell) \in \mathbb{Z}^2$, with site positions

$$\boxed{\mathbf{r}_A(n_1,\ell) = n_1\mathbf{R}_1 + \ell\mathbf{R}_2, \qquad \mathbf{r}_B(n_1,\ell) = \mathbf{r}_A(n_1,\ell) + \mathbf{s}_x.} \tag{13}$$

Fig. 1 shows the unit cell and the NN/Bravais vectors.

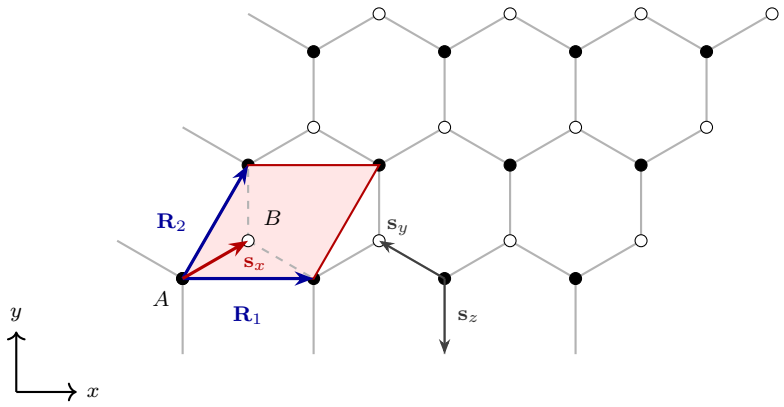


Figure 1: Honeycomb lattice. Filled circles: A; open circles: B. Intra-cell A–B offset is $\mathbf{s}_x$ (red); Bravais vectors $\mathbf{R}_1, \mathbf{R}_2$ in blue; the shaded rhombus is the unit cell, with its interior lattice bonds shown as grey dashes. Reference NN vectors $\mathbf{s}_z$ and $\mathbf{s}_y$ are shown in grey from a second A site.

The Fourier phase of an A–B bilinear is determined by the unit-cell label of the B operator, not directly by the physical A-to-B bond vector. This is the only point where some care is needed. By convention, the A site in cell $\mathbf{r}$ is placed at $\mathbf{r}$, while the B site in the same cell is placed at $\mathbf{r}+\mathbf{s}_x$. Hence the bilinear

$$ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}) \tag{14}$$

connects the site at $\mathbf{r}$ to the site at $\mathbf{r}+\boldsymbol{\Delta}+\mathbf{s}_x$. Its actual real-space connector is therefore

$$(\mathbf{r}+\boldsymbol{\Delta}+\mathbf{s}_x)-\mathbf{r}=\boldsymbol{\Delta}+\mathbf{s}_x. \tag{15}$$

If this bilinear is to represent the physical α-bond, this actual connector must be the NN vector $\mathbf{s}_\alpha$. Therefore

$$\boldsymbol{\Delta}_\alpha+\mathbf{s}_x=\mathbf{s}_\alpha, \qquad \boldsymbol{\Delta}_\alpha=\mathbf{s}_\alpha-\mathbf{s}_x. \tag{16}$$

This is not an additional geometrical assumption; it is only the conversion from absolute site positions to unit-cell labels. Since the x-bond was chosen as the intra-cell A–B bond, $\boldsymbol{\Delta}_x=0$, while the other two nearest-neighbor bonds connect A in cell $\mathbf{r}$ to B sites in neighboring cells:

$$\begin{aligned}
\boldsymbol{\Delta}_x &= \mathbf{s}_x-\mathbf{s}_x=\mathbf{0},\\
\boldsymbol{\Delta}_y &= \mathbf{s}_y-\mathbf{s}_x=(-\sqrt{3},0)=-\mathbf{R}_1,\\
\boldsymbol{\Delta}_z &= \mathbf{s}_z-\mathbf{s}_x=\left(-\frac{\sqrt{3}}{2},-\frac{3}{2}\right)=-\mathbf{R}_2.
\end{aligned} \tag{17}$$

Thus the Bloch phases are fixed once the unit cell is chosen. They are not chosen later in momentum space. The NN A–B bilinears used below are therefore

Bond	bilinear	actual A-to-B connector
x	$ic_A(\mathbf{r})c_B(\mathbf{r})$	$\mathbf{s}_x$
y	$ic_A(\mathbf{r})c_B(\mathbf{r}-\mathbf{R}_1)$	$\mathbf{s}_y$
z	$ic_A(\mathbf{r})c_B(\mathbf{r}-\mathbf{R}_2)$	$\mathbf{s}_z$

For same-sublattice bilinears no intra-cell A–B offset is involved, so the unit-cell displacement and the actual connector are the same. The Channel-2 mass terms use

Parameter	bilinear direction	displacement
Q_z	A–A or B–B along $\mathbf{p}_z$	$\mathbf{p}_z=\mathbf{R}_1$
Q_x	A–A or B–B along $\mathbf{p}_x$	$\mathbf{p}_x=\mathbf{R}_1-\mathbf{R}_2$
Q_y	A–A or B–B along $\mathbf{p}_y$	$\mathbf{p}_y=\mathbf{R}_2$

For the 3NN A–B bilinears the same unit-cell rule as above applies. The B operator is in cell $\mathbf{r}+\boldsymbol{\Delta}$, so the actual connector is $\boldsymbol{\Delta}+\mathbf{s}_x$:

Parameter	cell displacement in $c_B(\mathbf{r}+\boldsymbol{\Delta})$	actual connector
S_1	$\mathbf{R}_1-\mathbf{R}_2$	$(\sqrt{3},\,-1)$
S_2	$\mathbf{R}_2-\mathbf{R}_1$	$(0,\,2)$
S_3	$-\mathbf{R}_1-\mathbf{R}_2$	$(-\sqrt{3},\,-1)$

The actual connector is useful when drawing the ribbon geometry (Section 5); the bulk Fourier phase uses the cell displacement appearing in the operator label.

1.3 Mean-field channels

For each bond type α, the interaction couples two parallel α-bonds. The second bond is displaced from the first one by

$$\mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2, \qquad \mathbf{p}_y = \mathbf{R}_2, \qquad \mathbf{p}_z = \mathbf{R}_1. \tag{18}$$

These are Bravais displacements between unit cells, not physical A–B bond vectors. They are the NNN vectors of each triangular sublattice and will also be the directions of the same-sublattice mass channel. The three $\mathbf{p}_\alpha$ form a C_3 cycle:

$$C_3\mathbf{R}_1 = -\mathbf{R}_1 + \mathbf{R}_2, \qquad C_3\mathbf{R}_2 = -\mathbf{R}_1. \tag{19}$$

Combining $\mathbf{p}_\alpha$ with the A–B cell offsets $\boldsymbol{\Delta}_x = \mathbf{0}$, $\boldsymbol{\Delta}_y = -\mathbf{R}_1$, and $\boldsymbol{\Delta}_z = -\mathbf{R}_2$, the three bond-pair terms are explicitly:

$$H_J^{(x)} = J\sum_{\mathbf{r}} \big(ic_A(\mathbf{r})\, c_B(\mathbf{r})\big)\, \big(ic_A(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\, c_B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\big), \tag{20}$$

$$H_J^{(y)} = J\sum_{\mathbf{r}} \big(ic_A(\mathbf{r})\, c_B(\mathbf{r}-\mathbf{R}_1)\big)\, \big(ic_A(\mathbf{r}+\mathbf{R}_2)\, c_B(\mathbf{r}+\mathbf{R}_2-\mathbf{R}_1)\big), \tag{21}$$

$$H_J^{(z)} = J\sum_{\mathbf{r}} \big(ic_A(\mathbf{r})\, c_B(\mathbf{r}-\mathbf{R}_2)\big)\, \big(ic_A(\mathbf{r}+\mathbf{R}_1)\, c_B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\big). \tag{22}$$

Writing each $H_J^{(\alpha)} = J\,(ic_jc_k)\,(ic_\ell c_m)$:

Bond	j	k	ℓ	m
x	$A(\mathbf{r})$	$B(\mathbf{r})$	$A(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)$	$B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)$
y	$A(\mathbf{r})$	$B(\mathbf{r}-\mathbf{R}_1)$	$A(\mathbf{r}+\mathbf{R}_2)$	$B(\mathbf{r}+\mathbf{R}_2-\mathbf{R}_1)$
z	$A(\mathbf{r})$	$B(\mathbf{r}-\mathbf{R}_2)$	$A(\mathbf{r}+\mathbf{R}_1)$	$B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)$

The four sites j, k, ℓ, m are pairwise distinct for every α.

The same information can be summarized as

$$H_J^{(\alpha)} = J\sum_{\mathbf{r}} \Big[ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\Big]\Big[ic_A(\mathbf{r}+\mathbf{p}_\alpha)c_B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha)\Big]. \tag{23}$$

For example, in the z-bond case $\boldsymbol{\Delta}_z = -\mathbf{R}_2$ and $\mathbf{p}_z = \mathbf{R}_1$. Hence the second bond is

$$A(\mathbf{r}+\mathbf{R}_1)\ --\ B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2), \tag{24}$$

which gives Eq. (22). The x and y cases are obtained by the same substitution. This is the only geometric input used in the mean-field decoupling.

Channel structure from Majorana reordering. The site assignment $(j,k,\ell,m) = (A,B,A,B)$ is the same for every bond type. Write

$$B_{ab} = ic_a c_b. \tag{25}$$

Then the original quartic factor is $B_{jk}B_{\ell m}$. The relative signs of the three mean-field channels can be checked directly from Majorana anticommutation:

$$B_{jk}B_{\ell m} = (ic_j c_k)(ic_\ell c_m) = -c_j c_k c_\ell c_m, \tag{26}$$

whereas

$$B_{j\ell}B_{km} = (ic_j c_\ell)(ic_k c_m) = -c_j c_\ell c_k c_m = +c_j c_k c_\ell c_m = -B_{jk}B_{\ell m}, \tag{27}$$

and

$$B_{jm}B_{k\ell} = (ic_j c_m)(ic_k c_\ell) = -c_j c_m c_k c_\ell = -c_j c_k c_\ell c_m = B_{jk}B_{\ell m}. \tag{28}$$

Thus the direct, exchange, and cross pairings carry coefficients $+1, -1, +1$, respectively, relative to the common factor J. Applying this rearrangement to $H_J^{(\alpha)}$ produces three terms, one per channel:

- Channel 1 (direct, coefficient $+J$): $(jk)(\ell m) = (AB)(AB)$, an inter-sublattice NN bilinear with mean-field parameter $P_\alpha = \langle ic_A(\mathbf{r})\, c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\rangle$.
- Channel 2 (exchange, coefficient $-J$): $(j\ell)(km) = (AA)(BB)$, intra-sublattice NNN bilinears along $\mathbf{p}_\alpha$ with parameters $Q_{A,\alpha}$ and $Q_{B,\alpha}$ (Section 4.3).
- Channel 3 (cross, coefficient $+J$): $(jm)(k\ell) = (AB)(BA)$, inter-sublattice 3NN bilinears at cell displacements $\mathbf{p}_\alpha \pm \boldsymbol{\Delta}_\alpha$ with parameters S_1, S_2, S_3 (Section 4.4).

The main derivation proceeds in Section 2, where the full mean-field kernel is obtained directly from the real-space Hamiltonian, the mean-field decoupling, and the Fourier convention. The older channel-by-channel derivations are kept only as consistency checks in Section 4; they are not used as independent input for the final formulas.

2 Mean-field kernel

We now derive the two-band Majorana kernel from the real-space bond-pair Hamiltonian. The channel-by-channel derivations in Section 4 are used only as checks. The quadratic kernel in the (c_A, c_B) basis is

$$\boxed{\mathcal{H}(\mathbf{k}) = \begin{pmatrix} -m(\mathbf{k}) & i\,\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & +m(\mathbf{k}) \end{pmatrix}, \qquad \omega(\mathbf{k}) = \sqrt{m(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2},} \tag{29}$$

with eigenvalues $\pm\omega(\mathbf{k})$.

2.1 General kernel

With only translation symmetry, the mean-field theory has nine real parameters,

$$P_x, P_y, P_z; \qquad Q_x, Q_y, Q_z; \qquad S_1, S_2, S_3. \tag{30}$$

The purpose of this subsection is to obtain the functions $\Phi(\mathbf{k})$ and $m(\mathbf{k})$ in (29) directly from real space. The derivation uses only the bond-pair Hamiltonian, the mean-field decoupling, and the Fourier convention.

The flux-free bond-pair Hamiltonian is

$$H_J = J \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} \Big[ic_A(\mathbf{r}) c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha) \Big] \Big[ic_A(\mathbf{r}+\mathbf{p}_\alpha) c_B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha) \Big]. \tag{31}$$

Here

$$\boldsymbol{\Delta}_x = \mathbf{0}, \qquad \boldsymbol{\Delta}_y = -\mathbf{R}_1, \qquad \boldsymbol{\Delta}_z = -\mathbf{R}_2, \tag{32}$$

are the A-to-B cell displacements, while

$$\mathbf{p}_z = \mathbf{R}_1, \qquad \mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2, \qquad \mathbf{p}_y = \mathbf{R}_2 \tag{33}$$

are the separations between the two parallel bonds in a given bond pair. For a fixed α, it is useful to name the four Majoranas as

$$j = A(\mathbf{r}), \qquad k = B(\mathbf{r}+\boldsymbol{\Delta}_\alpha), \qquad \ell = A(\mathbf{r}+\mathbf{p}_\alpha), \qquad m = B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha). \tag{34}$$

Then the local factor is $J(ic_j c_k)(ic_\ell c_m)$. Moving the Majorana operators into the desired bilinear pairings gives

$$(jk)(\ell m), \qquad -(j\ell)(km), \qquad (jm)(k\ell), \tag{35}$$

namely the direct, exchange, and cross channels with relative signs $+, -, +$.

We use

$$c_X(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} c_X(\mathbf{k}), \qquad X = A, B. \tag{36}$$

Before decoupling the three channels, we first translate the elementary real-space bilinears into momentum space. Here δ is always a cell displacement. The point of the calculation is to keep track of which minus signs come from a Bloch phase and which minus signs come from swapping Majorana operators.

The A–B part of the kernel is read from the coefficient of $ic_A(-\mathbf{k})c_B(\mathbf{k})$. For

$$H_{AB}(\delta) = \lambda \sum_{\mathbf{r}} ic_A(\mathbf{r}) c_B(\mathbf{r}+\delta), \tag{37}$$

the Fourier transform gives

$$\begin{aligned} H_{AB}(\delta) &= \frac{i\lambda}{N} \sum_{\mathbf{r},\mathbf{k},\mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{r}} e^{i\mathbf{k}'\cdot(\mathbf{r}+\delta)} c_A(\mathbf{k}) c_B(\mathbf{k}') \\ &= i\lambda \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\delta} c_A(\mathbf{k}) c_B(-\mathbf{k}) \\ &= i\lambda \sum_{\mathbf{k}} e^{+i\mathbf{k}\cdot\delta} c_A(-\mathbf{k}) c_B(\mathbf{k}). \end{aligned} \tag{38}$$

Comparing this with the off-diagonal term $i\Phi(\mathbf{k})c_A(-\mathbf{k})c_B(\mathbf{k})$, one obtains

$$\Phi_{AB}(\mathbf{k};\delta,\lambda) = \lambda e^{i\mathbf{k}\cdot\delta}. \tag{39}$$

For a B–A bilinear the extra sign is not a Bloch phase. It is the Majorana swap needed to restore A–B order:

$$\begin{aligned}\lambda\sum_{\mathbf{r}} ic_B(\mathbf{r})c_A(\mathbf{r}+\delta) &= -\lambda\sum_{\mathbf{r}} ic_A(\mathbf{r}+\delta)c_B(\mathbf{r}) \\ &= -\lambda\sum_{\mathbf{r}'} ic_A(\mathbf{r}')c_B(\mathbf{r}'-\delta).\end{aligned} \tag{40}$$

This is the previous A–B result with displacement $-\delta$ and coefficient $-\lambda$. Therefore

$$\Phi_{BA}(\mathbf{k};\delta,\lambda) = -\lambda e^{-i\mathbf{k}\cdot\delta}. \tag{41}$$

For a same-sublattice hopping, it is better not to introduce a new $+\delta$ and $-\delta$ hopping parameter. The opposite orientation is already contained in the same Majorana bilinear. Start from the forward operator

$$H_X(\delta) = \lambda\sum_{\mathbf{r}} ic_X(\mathbf{r})c_X(\mathbf{r}+\delta), \qquad \lambda\in\mathbb{R}. \tag{42}$$

The same Hermitian operator may also be read in the reversed order because

$$ic_X(\mathbf{r})c_X(\mathbf{r}+\delta) = -ic_X(\mathbf{r}+\delta)c_X(\mathbf{r}). \tag{43}$$

Thus, in the kernel, the forward reading contributes $+i\lambda e^{+i\mathbf{k}\cdot\delta}$, whereas the reversed reading contributes $-i\lambda e^{-i\mathbf{k}\cdot\delta}$. The diagonal entry is the sum of these two oriented readings:

$$\begin{aligned}[\mathcal{H}_X(\mathbf{k})]_{XX} &= i\lambda e^{+i\mathbf{k}\cdot\delta} - i\lambda e^{-i\mathbf{k}\cdot\delta} \\ &= i\lambda\left(e^{+i\mathbf{k}\cdot\delta} - e^{-i\mathbf{k}\cdot\delta}\right) = -2\lambda\sin(\mathbf{k}\cdot\delta).\end{aligned} \tag{44}$$

The sine factor is therefore forced by the antisymmetry of an oriented same-sublattice Majorana hopping. It is not an onsite mass and it is not an extra momentum-space assumption.

Direct channel. The direct mean field is

$$P_\alpha \equiv \left\langle ic_A(\mathbf{r})c_B(\mathbf{r}+\mathbf{\Delta}_\alpha)\right\rangle. \tag{45}$$

For $X = ic_jc_k$ and $Y = ic_\ell c_m$,

$$XY = (X-P_\alpha)(Y-P_\alpha) + P_\alpha X + P_\alpha Y - P_\alpha^2. \tag{46}$$

Dropping the quadratic fluctuation gives

$$(ic_jc_k)(ic_\ell c_m) \longrightarrow P_\alpha(ic_\ell c_m) + P_\alpha(ic_jc_k) - P_\alpha^2. \tag{47}$$

The two operator terms are the same A–B bilinear after translating $\mathbf{r}\mapsto\mathbf{r}-\mathbf{p}_\alpha$ in the first one. Therefore

$$H_P = 2J\sum_{\alpha,\mathbf{r}} P_\alpha ic_A(\mathbf{r})c_B(\mathbf{r}+\mathbf{\Delta}_\alpha) - NJ\sum_\alpha P_\alpha^2. \tag{48}$$

Including the bare Kitaev hopping, the direct contribution to the off-diagonal kernel is

$$\Phi_P(\mathbf{k}) = \sum_{\alpha=x,y,z} (K_1 + 2JP_\alpha) e^{i\mathbf{k}\cdot\boldsymbol{\Delta}_\alpha}. \tag{49}$$

Equivalently,

$$\Phi_P(\mathbf{k}) = (K_1 + 2JP_x) + (K_1 + 2JP_y) e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K_1 + 2JP_z) e^{-i\mathbf{k}\cdot\mathbf{R}_2}. \tag{50}$$

Exchange channel. The exchange pairing connects the two A sites and the two B sites:

$$Q_{A,\alpha} \equiv \big\langle ic_A(\mathbf{r}) c_A(\mathbf{r}+\mathbf{p}_\alpha) \big\rangle, \qquad Q_{B,\alpha} \equiv \big\langle ic_B(\mathbf{r}) c_B(\mathbf{r}+\mathbf{p}_\alpha) \big\rangle. \tag{51}$$

Because the exchange channel has the Fierz coefficient $-J$,

$$-(ic_j c_\ell)(ic_k c_m) \longrightarrow -\Big[Q_{A,\alpha}(ic_k c_m) + Q_{B,\alpha}(ic_j c_\ell) - Q_{A,\alpha} Q_{B,\alpha} \Big]. \tag{52}$$

Thus the A–A bilinear carries coefficient $-JQ_{B,\alpha}$, and the B–B bilinear carries coefficient $-JQ_{A,\alpha}$. Applying (44) gives

$$[\mathcal{H}_Q(\mathbf{k})]_{AA} = 2J \sum_\alpha Q_{B,\alpha} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \qquad [\mathcal{H}_Q(\mathbf{k})]_{BB} = 2J \sum_\alpha Q_{A,\alpha} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{53}$$

Now separate the average and staggered combinations,

$$Q_{+,\alpha} = \frac{Q_{A,\alpha} + Q_{B,\alpha}}{2}, \qquad Q_{-,\alpha} = \frac{Q_{A,\alpha} - Q_{B,\alpha}}{2}. \tag{54}$$

The $Q_{+,\alpha}$ part is proportional to the identity matrix and to the odd function $\sin(\mathbf{k}\cdot\mathbf{p}_\alpha)$. Its contribution to the filled-band Brillouin-zone sum vanishes by $\mathbf{k} \leftrightarrow -\mathbf{k}$. The constant term, however, is

$$JNQ_{A,\alpha}Q_{B,\alpha} = JN(Q_{+,\alpha}^2 - Q_{-,\alpha}^2). \tag{55}$$

Therefore stationarity gives

$$\frac{\partial E_{\mathrm{GS}}}{\partial Q_{+,\alpha}} = 2JNQ_{+,\alpha} = 0, \tag{56}$$

so that $Q_{+,\alpha} = 0$. We may therefore write

$$Q_{A,\alpha} = Q_\alpha, \qquad Q_{B,\alpha} = -Q_\alpha. \tag{57}$$

Then (53) becomes

$$[\mathcal{H}_Q(\mathbf{k})]_{AA} = -m_J(\mathbf{k}), \qquad [\mathcal{H}_Q(\mathbf{k})]_{BB} = +m_J(\mathbf{k}), \tag{58}$$

where

$$m_J(\mathbf{k}) = 2J \sum_{\alpha=x,y,z} Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{59}$$

Using the ordering in (33),

$$m_J(\mathbf{k}) = 2J\Big[Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\big(\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\big) + Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2) \Big]. \tag{60}$$

Diagnostic mass. The additional chiral NNN Majorana hopping is not varied in the SCF loop. It is inserted after the $\kappa = 0$ fixed point has been found. Its orientation is derived in Subsection 2.2; with $(\nu_z, \nu_x, \nu_y) = (+1, -1, -1)$, the same same-sublattice Fourier rule above gives

$$m_\kappa(\mathbf{k}) = 2\kappa\Big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\Big]. \tag{61}$$

The full diagonal mass is $m(\mathbf{k}) = m_J(\mathbf{k}) + m_\kappa(\mathbf{k})$.

Cross channel. The cross pairing consists of one A–B bilinear and one B–A bilinear. We use the three A–B expectation values

$$\begin{aligned} S_1 &\equiv \big\langle ic_A(\mathbf{r})c_B(\mathbf{r} + \mathbf{R}_1 - \mathbf{R}_2)\big\rangle, \\ S_2 &\equiv \big\langle ic_A(\mathbf{r})c_B(\mathbf{r} + \mathbf{R}_2 - \mathbf{R}_1)\big\rangle, \\ S_3 &\equiv \big\langle ic_A(\mathbf{r})c_B(\mathbf{r} - \mathbf{R}_1 - \mathbf{R}_2)\big\rangle. \end{aligned} \tag{62}$$

A B–A expectation value is converted to this A–B list by translation and one Majorana swap:

$$\begin{aligned} \big\langle ic_B(\mathbf{r})c_A(\mathbf{r} + \delta)\big\rangle &= \big\langle ic_B(\mathbf{r} - \delta)c_A(\mathbf{r})\big\rangle \\ &= -\big\langle ic_A(\mathbf{r})c_B(\mathbf{r} - \delta)\big\rangle. \end{aligned} \tag{63}$$

The cross decoupling is

$$(ic_jc_m)(ic_kc_\ell) \longrightarrow \langle ic_jc_m\rangle(ic_kc_\ell) + (ic_jc_m)\langle ic_kc_\ell\rangle - \langle ic_jc_m\rangle\langle ic_kc_\ell\rangle. \tag{64}$$

The two residual operator displacements are

$$(j,m): \quad \mathbf{p}_\alpha + \boldsymbol{\Delta}_\alpha, \qquad (k,\ell): \quad \mathbf{p}_\alpha - \boldsymbol{\Delta}_\alpha. \tag{65}$$

Reading the two expectation values in each row gives

α	$\mathbf{p}_\alpha + \boldsymbol{\Delta}_\alpha$	$\langle ic_jc_m\rangle$	$\mathbf{p}_\alpha - \boldsymbol{\Delta}_\alpha$	$\langle ic_kc_\ell\rangle$
x	$\mathbf{R}_1 - \mathbf{R}_2$	S_1	$\mathbf{R}_1 - \mathbf{R}_2$	$-S_2$
y	$\mathbf{R}_2 - \mathbf{R}_1$	S_2	$\mathbf{R}_1 + \mathbf{R}_2$	$-S_3$
z	$\mathbf{R}_1 - \mathbf{R}_2$	S_1	$\mathbf{R}_1 + \mathbf{R}_2$	$-S_3$

The first term in (64) leaves the B–A operator (k,ℓ) and is therefore read with (41). The second term leaves the A–B operator (j,m) and is read with (39). Thus the six contributions to $\Phi(\mathbf{k})$ are

α	B–A contribution	A–B contribution
x	$-JS_1e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$	$-JS_2e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$
y	$-JS_2e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}$	$-JS_3e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$
z	$-JS_1e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}$	$-JS_3e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$

Collecting equal phase factors gives

$$\Delta\Phi_S(\mathbf{k}) = -J(S_2+S_3)e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} - J(S_1+S_3)e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} - J(S_1+S_2)e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}. \tag{66}$$

The constant from the cross channel is

$$NJ(S_1S_2 + S_1S_3 + S_2S_3), \tag{67}$$

where each product appears once after summing the three bond-pair types.

Adding (50) and (66),

$$\boxed{\begin{aligned}\Phi(\mathbf{k}) = (K_1 + 2JP_x) + (K_1 + 2JP_y)\, e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K_1 + 2JP_z)\, e^{-i\mathbf{k}\cdot\mathbf{R}_2} \\ - J(S_2{+}S_3)\, e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} - J(S_1{+}S_3)\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} - J(S_1{+}S_2)\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}.\end{aligned}} \tag{68}$$

Combining (60) and (61),

$$\boxed{\begin{aligned} m(\mathbf{k}) &= m_J(\mathbf{k}) + m_\kappa(\mathbf{k}), \\ m_J(\mathbf{k}) &= 2J\big[\, Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) + Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2)\,\big], \\ m_\kappa(\mathbf{k}) &= 2\kappa\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\,\big]. \end{aligned}} \tag{69}$$

The corresponding total ground-state energy is

$$\boxed{E_{\rm GS} = -NJ\sum_\alpha P_\alpha^2 - NJ\sum_\alpha Q_\alpha^2 + NJ\big(S_1S_2 + S_1S_3 + S_2S_3\big) - \sum_{\mathbf{k}\in 1\rm BZ} \omega(\mathbf{k};\kappa).} \tag{70}$$

Here $E_{\rm GS}$ is the total energy for N unit cells, and the Brillouin-zone sum contains $N_k = N$ momenta. The code reports the energy per unit cell. Notice that m_J and m_κ are structurally different: m_J has a common plus sign in the three directions and obtains its chirality from the values of Q_α, whereas m_κ has the fixed closed-loop signs $(+,-,-)$. They can be merged into a single coefficient only after the C_3 reduction in (129).

2.2 Oriented Q and κ

The mass sector contains two same-sublattice Majorana hoppings. One is the interaction-generated exchange channel, denoted by Q_α. The other is an externally inserted diagnostic hopping, denoted by κ. They enter the same part of the two-band kernel because both are staggered NNN hoppings: the A and B sublattices carry opposite diagonal entries. This subsection fixes the real-space arrow convention first, and only then writes the corresponding sine form factors in momentum space.

For a Majorana bilinear, an oriented arrow and the same link with the opposite arrow differ by a minus sign. For one sublattice,

$$\begin{aligned}\langle ic_X(\mathbf{r})c_X(\mathbf{r}-\mathbf{p})\rangle &= \langle ic_X(\mathbf{r}+\mathbf{p})c_X(\mathbf{r})\rangle \\ &= -\langle ic_X(\mathbf{r})c_X(\mathbf{r}+\mathbf{p})\rangle\,, \qquad X = A, B.\end{aligned} \tag{71}$$

The coefficient attached to a NNN link is therefore not attached to an unoriented segment. It is attached to a directed arrow.

The positive reference arrows used throughout the note are

$$\mathbf{p}_z = \mathbf{R}_1, \qquad \mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2, \qquad \mathbf{p}_y = \mathbf{R}_2. \tag{72}$$

Equivalently, Q_z is the coefficient attached to the arrow $+\mathbf{p}_z$, Q_x to $+\mathbf{p}_x$, and Q_y to $+\mathbf{p}_y$. By (71), the coefficient attached to the reversed arrow $-\mathbf{p}_\alpha$ is $-Q_\alpha$.

Now rotate the lattice by $+120^\circ$ about a sublattice site. The Bravais vectors transform as

$$C_3\mathbf{R}_1 = \mathbf{R}_2 - \mathbf{R}_1, \qquad C_3\mathbf{R}_2 = -\mathbf{R}_1. \tag{73}$$

Thus the three positive reference arrows transform as

$$C_3\mathbf{p}_z = -\mathbf{p}_x, \qquad C_3\mathbf{p}_x = +\mathbf{p}_y, \qquad C_3\mathbf{p}_y = -\mathbf{p}_z. \tag{74}$$

The minus signs in (74) are geometric signs: after rotation, the corresponding arrow points opposite to the positive reference arrow chosen in (72).

Imposing C_3 symmetry means that the coefficient on a directed arrow equals the coefficient on its rotated directed arrow. Therefore

$$\begin{aligned} Q_z &= \mathrm{coeff}(+\mathbf{p}_z) = \mathrm{coeff}(C_3\mathbf{p}_z) = \mathrm{coeff}(-\mathbf{p}_x) = -Q_x, \\ Q_x &= \mathrm{coeff}(+\mathbf{p}_x) = \mathrm{coeff}(C_3\mathbf{p}_x) = \mathrm{coeff}(+\mathbf{p}_y) = +Q_y, \\ Q_y &= \mathrm{coeff}(+\mathbf{p}_y) = \mathrm{coeff}(C_3\mathbf{p}_y) = \mathrm{coeff}(-\mathbf{p}_z) = -Q_z. \end{aligned} \tag{75}$$

Solving these three relations gives

$$(Q_z, Q_x, Q_y) = (Q, -Q, -Q). \tag{76}$$

This is not an arbitrary choice made in momentum space. The relative signs are fixed once the positive arrows $(\mathbf{p}_z, \mathbf{p}_x, \mathbf{p}_y)$ are fixed. Only the overall sign of Q remains a branch choice: replacing $Q \to -Q$ reverses all three oriented coefficients.

Next we explain why this oriented hopping appears as a mass term. Consider one same-sublattice hopping with amplitude λ along a directed displacement δ,

$$H_X(\delta) = \lambda \sum_{\mathbf{r}} i c_X(\mathbf{r}) c_X(\mathbf{r} + \delta). \tag{77}$$

The opposite arrow is not an independent term; it has amplitude $-\lambda$ by Majorana antisymmetry. Fourier transforming the two orientations gives

$$\lambda e^{i\mathbf{k}\cdot\delta} - \lambda e^{-i\mathbf{k}\cdot\delta} = 2i\lambda \sin(\mathbf{k}\cdot\delta), \tag{78}$$

or, in the matrix convention used here,

$$\lambda \sum_{\mathbf{r}} i c_X(\mathbf{r}) c_X(\mathbf{r} + \delta) \quad \Longrightarrow \quad [\mathcal{H}_\lambda(\mathbf{k})]_{XX} = -2\lambda \sin(\mathbf{k}\cdot\delta). \tag{79}$$

The exchange channel has the staggered real-space form

$$H_Q = J \sum_{\mathbf{r}} \sum_{\alpha=z,x,y} Q_\alpha \left[i c_A(\mathbf{r}) c_A(\mathbf{r} + \mathbf{p}_\alpha) - i c_B(\mathbf{r}) c_B(\mathbf{r} + \mathbf{p}_\alpha) \right]. \tag{80}$$

For the A sublattice the amplitude in (79) is $\lambda = JQ_\alpha$; for the B sublattice it is $\lambda = -JQ_\alpha$. Hence

$$[\mathcal{H}_Q(\mathbf{k})]_{AA} = -2J \sum_{\alpha=z,x,y} Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \qquad [\mathcal{H}_Q(\mathbf{k})]_{BB} = +2J \sum_{\alpha=z,x,y} Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{81}$$

Comparing with

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} -m(\mathbf{k}) & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & m(\mathbf{k}) \end{pmatrix}, \tag{82}$$

we identify

$$m_J(\mathbf{k}) = 2J \sum_{\alpha=z,x,y} Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{83}$$

In explicit Bravais-vector form,

$$m_J(\mathbf{k}) = 2J \left[Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\big(\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\big) + Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2) \right]. \tag{84}$$

With the C_3 pattern (76), this reduces to

$$m_J^{C_3}(\mathbf{k}) = 2JQ\, M_0(\mathbf{k}), \qquad M_0(\mathbf{k}) \equiv \sin(\mathbf{k}\cdot\mathbf{p}_z) - \sin(\mathbf{k}\cdot\mathbf{p}_x) - \sin(\mathbf{k}\cdot\mathbf{p}_y). \tag{85}$$

This last equation is often the most economical way to remember the sign structure: the sign pattern is already fixed in real space, while the Fourier transform supplies the sine functions.

The diagnostic hopping κ is chosen to have a fixed chiral circulation on each triangular sublattice. The positive reference vectors obey

$$\mathbf{p}_z - \mathbf{p}_x - \mathbf{p}_y = 0. \tag{86}$$

Thus the directed triangular path

$$\mathbf{r} \longrightarrow \mathbf{r} + \mathbf{p}_z \longrightarrow \mathbf{r} + \mathbf{p}_y \longrightarrow \mathbf{r} \tag{87}$$

is closed. Its three directed steps are

$$+\mathbf{p}_z, \qquad (\mathbf{r} + \mathbf{p}_y) - (\mathbf{r} + \mathbf{p}_z) = \mathbf{p}_y - \mathbf{p}_z = -\mathbf{p}_x, \qquad -\mathbf{p}_y. \tag{88}$$

Therefore, relative to the fixed reference arrows, the chiral coefficients are

$$(\nu_z, \nu_x, \nu_y) = (+1, -1, -1). \tag{89}$$

The corresponding post-SCF hopping is

$$H_\kappa = \kappa \sum_{\mathbf{r}} \sum_{\alpha=z,x,y} \nu_\alpha \left[ic_A(\mathbf{r}) c_A(\mathbf{r} + \mathbf{p}_\alpha) - ic_B(\mathbf{r}) c_B(\mathbf{r} + \mathbf{p}_\alpha) \right]. \tag{90}$$

This has exactly the same staggered diagonal structure as H_Q, with JQ_α replaced by $\kappa\nu_\alpha$. Therefore it contributes to the same mass component:

$$m_\kappa(\mathbf{k}) = 2\kappa \sum_{\alpha=z,x,y} \nu_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{91}$$

Using (89),

$$m_\kappa(\mathbf{k}) = 2\kappa \left[\sin(\mathbf{k}\cdot\mathbf{p}_z) - \sin(\mathbf{k}\cdot\mathbf{p}_x) - \sin(\mathbf{k}\cdot\mathbf{p}_y) \right] = 2\kappa M_0(\mathbf{k}). \tag{92}$$

This explains why κ is added to the m term rather than to $\Phi(\mathbf{k})$: it is a same-sublattice staggered hopping, hence a diagonal sublattice-odd term in the matrix convention (82). Since the quadratic Hamiltonian is linear in hopping amplitudes, the two mass contributions simply add,

$$m(\mathbf{k}) = m_J(\mathbf{k}) + m_\kappa(\mathbf{k}) = 2 \sum_{\alpha=z,x,y} (JQ_\alpha + \kappa\nu_\alpha) \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{93}$$

In the C_3-symmetric sector this becomes

$$m^{C_3}(\mathbf{k}) = 2(JQ + \kappa) M_0(\mathbf{k}). \tag{94}$$

Thus $JQ \to JQ+\kappa$ is only a shorthand valid after the C_3 reduction and with the present arrow convention.

Finally, the role of κ in this note is diagnostic. The SCF map is solved at $\kappa = 0$:

$$(P^*, Q^*, S^*) \quad \text{solves the SCF equations with} \quad m(\mathbf{k}) = m_J(\mathbf{k}). \tag{95}$$

After that fixed point is obtained, spectra and Chern numbers can be evaluated with

$$m_{\text{obs}}(\mathbf{k}) = m_J(\mathbf{k}; P^*, Q^*, S^*) + m_\kappa(\mathbf{k}). \tag{96}$$

Although $m_\kappa(\mathbf{k})$ is not a constant over the Brillouin zone, near a zero of the $\kappa = 0$ spectrum it acts as the local Dirac mass. If $\mathbf{k}_i$ is a gapless point and $\mathbf{k} = \mathbf{k}_i + \mathbf{q}$, then

$$m_\kappa(\mathbf{k}_i + \mathbf{q}) = m_\kappa(\mathbf{k}_i) + \mathbf{q}\cdot\nabla_{\mathbf{k}} m_\kappa(\mathbf{k}_i) + O(|\mathbf{q}|^2), \tag{97}$$

so the leading local mass at that cone is

$$M_i^\kappa = m_\kappa(\mathbf{k}_i). \tag{98}$$

The Chern number itself, however, is computed from the full lattice sine form factor, not from the constant approximation alone.

2.3 SCF update

This subsection derives the SCF equations in the form used in the numerical calculation. The input of one SCF step is a trial set of mean-field parameters inside the quadratic kernel. The output is obtained by evaluating the same real-space bilinears in the ground state of that quadratic Hamiltonian. A self-consistent solution is a fixed point of this map.

The diagnostic mass κ is not included in this map. It is added only after a zero-κ mean-field solution has been found. Thus the kernel used inside the SCF update is

$$\mathcal{H}_0(\mathbf{k}) = \begin{pmatrix} -m_J(\mathbf{k}) & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & m_J(\mathbf{k}) \end{pmatrix}, \qquad \omega_0(\mathbf{k}) = \sqrt{m_J(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2}. \tag{99}$$

At each momentum, the negative-energy state is occupied. Write its normalized eigenvector as

$$u_-(\mathbf{k}) = \begin{pmatrix} u_A(\mathbf{k}) \\ u_B(\mathbf{k}) \end{pmatrix}, \qquad \mathcal{H}_0(\mathbf{k}) u_-(\mathbf{k}) = -\omega_0(\mathbf{k}) u_-(\mathbf{k}). \tag{100}$$

We now solve this two-component equation directly. The two component equations are

$$\begin{aligned} -m_J(\mathbf{k})u_A(\mathbf{k}) + i\Phi(\mathbf{k})u_B(\mathbf{k}) &= -\omega_0(\mathbf{k})u_A(\mathbf{k}), \\ -i\Phi^*(\mathbf{k})u_A(\mathbf{k}) + m_J(\mathbf{k})u_B(\mathbf{k}) &= -\omega_0(\mathbf{k})u_B(\mathbf{k}). \end{aligned} \tag{101}$$

From the second line,

$$u_B(\mathbf{k}) = \frac{i\Phi^*(\mathbf{k})}{\omega_0(\mathbf{k}) + m_J(\mathbf{k})} u_A(\mathbf{k}). \tag{102}$$

This expression is only a convenient way to fix the relative phase. The final relations below are continuous even at special momenta where this particular ratio is ill-conditioned. Using

$$|\Phi(\mathbf{k})|^2 = \big[\omega_0(\mathbf{k}) - m_J(\mathbf{k})\big]\big[\omega_0(\mathbf{k}) + m_J(\mathbf{k})\big] \tag{103}$$

and $|u_A|^2 + |u_B|^2 = 1$, one obtains

$$|u_A(\mathbf{k})|^2 = \frac{\omega_0(\mathbf{k}) + m_J(\mathbf{k})}{2\omega_0(\mathbf{k})}, \qquad |u_B(\mathbf{k})|^2 = \frac{\omega_0(\mathbf{k}) - m_J(\mathbf{k})}{2\omega_0(\mathbf{k})}. \tag{104}$$

The SCF equations need only two combinations of the lower-band wavefunction. The A–B coherence is

$$u_A^*(\mathbf{k})u_B(\mathbf{k}) = \frac{i\Phi^*(\mathbf{k})}{2\omega_0(\mathbf{k})}, \tag{105}$$

and the sublattice imbalance is

$$|u_A(\mathbf{k})|^2 - |u_B(\mathbf{k})|^2 = \frac{m_J(\mathbf{k})}{\omega_0(\mathbf{k})}. \tag{106}$$

Thus the P and S updates are controlled by the A–B coherence, while the Q update is controlled by the difference between A- and B-sublattice weights. This is the main physical content of the SCF formula.

We next state the real-space bilinears whose expectation values define the output parameters. For the nearest-neighbor A–B channels,

$$\begin{aligned} P_x^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r} + \boldsymbol{\Delta}_x) \rangle\,, \quad \boldsymbol{\Delta}_x = \mathbf{0}, \\ P_y^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r} + \boldsymbol{\Delta}_y) \rangle\,, \quad \boldsymbol{\Delta}_y = -\mathbf{R}_1, \\ P_z^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r} + \boldsymbol{\Delta}_z) \rangle\,, \quad \boldsymbol{\Delta}_z = -\mathbf{R}_2. \end{aligned} \tag{107}$$

These three equations are written separately because they fix the actual operator convention. Once this is fixed, the update can be written with a single bond index $\alpha = x, y, z$.

For the same-sublattice channel, the order parameter is the half-difference between the A- and B-sublattice oriented correlators:

$$\begin{aligned} Q_z^{\text{out}} &= \frac{1}{2}\left[\langle i c_A(\mathbf{r}) c_A(\mathbf{r} + \mathbf{p}_z) \rangle - \langle i c_B(\mathbf{r}) c_B(\mathbf{r} + \mathbf{p}_z) \rangle\right], \\ Q_x^{\text{out}} &= \frac{1}{2}\left[\langle i c_A(\mathbf{r}) c_A(\mathbf{r} + \mathbf{p}_x) \rangle - \langle i c_B(\mathbf{r}) c_B(\mathbf{r} + \mathbf{p}_x) \rangle\right], \\ Q_y^{\text{out}} &= \frac{1}{2}\left[\langle i c_A(\mathbf{r}) c_A(\mathbf{r} + \mathbf{p}_y) \rangle - \langle i c_B(\mathbf{r}) c_B(\mathbf{r} + \mathbf{p}_y) \rangle\right]. \end{aligned} \tag{108}$$

The minus sign between the two sublattices is what makes this channel a sublattice mass rather than a common same-sublattice hopping.

The longer-range A–B channels are

$$\begin{aligned} S_1^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\rangle\,, \\ S_2^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r}-\mathbf{R}_1+\mathbf{R}_2)\rangle\,, \\ S_3^{\text{out}} &= \langle i c_A(\mathbf{r}) c_B(\mathbf{r}-\mathbf{R}_1-\mathbf{R}_2)\rangle\,. \end{aligned} \tag{109}$$

These definitions contain no momentum-space assumption. The phases and sine factors below follow only after Fourier transforming these real-space operators.

Use the Fourier convention

$$c_X(\mathbf{r}) = \frac{1}{\sqrt{N_k}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} c_X(\mathbf{k}), \qquad X = A, B. \tag{110}$$

Consider first a general A–B bilinear with cell displacement δ:

$$P_\delta \equiv \langle i c_A(\mathbf{r}) c_B(\mathbf{r}+\delta)\rangle\,. \tag{111}$$

After inserting (110), translation invariance fixes the two momenta to be opposite. The two equivalent Majorana contractions can be written as

$$\begin{aligned} P_\delta &= \frac{i}{N_k} \sum_{\mathbf{k}} \left[e^{i\mathbf{k}\cdot\delta} u_A^*(\mathbf{k}) u_B(\mathbf{k}) - e^{-i\mathbf{k}\cdot\delta} u_B^*(\mathbf{k}) u_A(\mathbf{k}) \right] \\ &= -\frac{2}{N_k} \sum_{\mathbf{k}} \mathrm{Im}\left[e^{i\mathbf{k}\cdot\delta} u_A^*(\mathbf{k}) u_B(\mathbf{k}) \right]. \end{aligned} \tag{112}$$

The second term is not an independent assumption; it is the Majorana partner of the first term and is needed to make the expectation value real. Now use (105). Since $-2\,\mathrm{Im}(iZ) = -2\,\mathrm{Re}\,Z$, we get

$$\langle i c_A(\mathbf{r}) c_B(\mathbf{r}+\delta)\rangle = -\frac{1}{N_k} \sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\delta} \Phi^*(\mathbf{k}) \right]}{\omega_0(\mathbf{k})}. \tag{113}$$

Putting $\delta = \boldsymbol{\Delta}_\alpha$ gives the nearest-neighbor update

$$P_\alpha^{\text{out}} = -\frac{1}{N_k} \sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\Delta}_\alpha} \Phi^*(\mathbf{k}) \right]}{\omega_0(\mathbf{k})}, \qquad \alpha = x, y, z. \tag{114}$$

Putting instead the three longer A–B displacements from (109) gives

$$\begin{aligned} S_1^{\text{out}} &= -\frac{1}{N_k} \sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} \Phi^*(\mathbf{k}) \right]}{\omega_0(\mathbf{k})}, \\ S_2^{\text{out}} &= -\frac{1}{N_k} \sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot(-\mathbf{R}_1+\mathbf{R}_2)} \Phi^*(\mathbf{k}) \right]}{\omega_0(\mathbf{k})}, \\ S_3^{\text{out}} &= -\frac{1}{N_k} \sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)} \Phi^*(\mathbf{k}) \right]}{\omega_0(\mathbf{k})}. \end{aligned} \tag{115}$$

Now consider an oriented same-sublattice bilinear. For one sublattice $X = A, B$ and one displacement $\mathbf{p}_\alpha$,

$$Q_{X,\alpha} \equiv \langle i c_X(\mathbf{r}) c_X(\mathbf{r}+\mathbf{p}_\alpha)\rangle\,. \tag{116}$$

The two opposite momenta combine into a sine factor:

$$\langle i c_X(\mathbf{r}) c_X(\mathbf{r}+\mathbf{p}_\alpha)\rangle = -\frac{2}{N_k} \sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha) |u_X(\mathbf{k})|^2, \qquad X = A, B. \tag{117}$$

This is the same Fourier mechanism as in the chiral κ term: an oriented Majorana hopping is odd under arrow reversal, and the odd part of the Bloch phase is a sine. Taking the sublattice-staggered half-difference in (108) gives

$$Q_\alpha^{\rm out} = -\frac{1}{N_k}\sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\left[|u_A(\mathbf{k})|^2 - |u_B(\mathbf{k})|^2\right]. \tag{118}$$

Using (106), this becomes

$$Q_\alpha^{\rm out} = -\frac{1}{N_k}\sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\frac{m_J(\mathbf{k})}{\omega_0(\mathbf{k})}, \qquad \alpha = z, x, y. \tag{119}$$

Thus the Q update contains no extra sign convention beyond the real-space choice of the oriented vectors $\mathbf{p}_\alpha$ and the A-minus-B definition in (108).

Combining (114), (119), and (115) gives the nine-dimensional SCF map. A fixed point satisfies

$$X^* = X^{\rm out}(X^*), \qquad X = (P_z, P_x, P_y, Q_z, Q_x, Q_y, S_1, S_2, S_3). \tag{120}$$

Numerically, this fixed point is reached by linear mixing,

$$X_{n+1} = \lambda X^{\rm out}(X_n) + (1-\lambda)X_n, \qquad 0 < \lambda \le 1. \tag{121}$$

The parameter κ is not used in (99)–(121); it is inserted only after the fixed point has been selected.

2.4 C_3 reduction

We now specialize the general kernel to the branch that preserves the threefold lattice rotation. This step is useful because the formulae become shorter, but it should not be read as a new assumption about the Fourier phases. The signs have already been fixed in real space in Secs. 2.1 and 2.2; here we only substitute the C_3-symmetric parameter pattern into those formulae.

For the A–B hopping channels there is no arrow reversal in the C_3 orbit, so the three nearest-neighbour parameters are equal. The three S channels are also identified. For the same-sublattice Q channel, however, the reference arrows are not cyclically permuted without signs. As shown in (74), a rotation sends some arrows to the opposite reference direction, and an oriented Majorana bilinear changes sign under arrow reversal. Therefore the C_3-symmetric reduction is

$$P_x = P_y = P_z \equiv P, \qquad (Q_z, Q_x, Q_y) = (Q, -Q, -Q), \qquad S_1 = S_2 = S_3 \equiv U. \tag{122}$$

The middle relation is the point that is easiest to misread. It does not mean that the physical state is less symmetric. It means that the three numbers are written relative to the fixed reference arrows $(\mathbf{p}_z, \mathbf{p}_x, \mathbf{p}_y)$, and those reference arrows themselves acquire signs under C_3.

Let

$$\tilde{K} \equiv K_1 + 2JP. \tag{123}$$

The nearest-neighbour part of (68) then becomes

$$(K_1 + 2JP_x) + (K_1 + 2JP_y)e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K_1 + 2JP_z)e^{-i\mathbf{k}\cdot\mathbf{R}_2} = \tilde{K}\left[1 + e^{-i\mathbf{k}\cdot\mathbf{R}_1} + e^{-i\mathbf{k}\cdot\mathbf{R}_2}\right]. \tag{124}$$

Similarly, each cross-channel coefficient in (68) contains the sum of two equal S parameters:

$$S_2 + S_3 = S_1 + S_3 = S_1 + S_2 = 2U. \tag{125}$$

Therefore

$$\boxed{\begin{aligned}\Phi(\mathbf{k})\big|_{C_3} &= \tilde{K}\big[1 + e^{-i\mathbf{k}\cdot\mathbf{R}_1} + e^{-i\mathbf{k}\cdot\mathbf{R}_2}\big] \\ &\quad - 2JU\big[e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}\big].\end{aligned}} \tag{126}$$

No additional sign convention is hidden in (126); the three third-neighbour phases are precisely the three phases already obtained from the cross-channel Fourier transform.

The mass term is where the oriented nature of Q and κ becomes most visible. Substituting (122) into the interaction-generated mass in (69) gives

$$\begin{aligned} m_J(\mathbf{k})\big|_{C_3} &= 2J\big[Q\sin(\mathbf{k}\cdot\mathbf{R}_1) - Q\sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - Q\sin(\mathbf{k}\cdot\mathbf{R}_2)\big] \\ &= 2JQ\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\big]. \end{aligned} \tag{127}$$

On the other hand, the diagnostic hopping already has the same closed-loop orientation:

$$m_\kappa(\mathbf{k}) = 2\kappa\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\big]. \tag{128}$$

Adding the two contributions yields

$$\boxed{m(\mathbf{k})\big|_{C_3} = 2(JQ+\kappa)\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\big].} \tag{129}$$

Thus the shorthand

$$JQ \longrightarrow JQ + \kappa \tag{130}$$

is valid only after the C_3 reduction and only because the Q pattern and the diagnostic κ hopping carry the same oriented sine form factor. It does not mean that κ is a momentum-independent mass, and it does not mean that κ is part of the SCF iteration.

The constant part of the energy also reduces directly. From (70),

$$\sum_\alpha P_\alpha^2 = 3P^2, \qquad \sum_\alpha Q_\alpha^2 = Q^2 + (-Q)^2 + (-Q)^2 = 3Q^2, \qquad S_1S_2 + S_1S_3 + S_2S_3 = 3U^2. \tag{131}$$

Therefore

$$\boxed{E_{\mathrm{GS}}\big|_{C_3} = -3NJ(P^2+Q^2-U^2) - \sum_{\mathbf{k}\in 1\mathrm{BZ}} \omega(\mathbf{k};\kappa).} \tag{132}$$

Here P, Q, U are selected self-consistently at $\kappa = 0$. After that, a nonzero κ changes the spectrum and the diagnostic Chern number through (129), but it does not change the SCF equations used to obtain P, Q, U.

In the noninteracting limit $J = 0$, the physical kernel no longer depends on P, Q, U because every mean-field correction is multiplied by J. The mass is then purely the post-SCF diagnostic hopping,

$$m(\mathbf{k})\big|_{J=0} = 2\kappa\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\big(\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)\big) - \sin(\mathbf{k}\cdot\mathbf{R}_2)\big]. \tag{133}$$

The sign of the resulting Chern number is evaluated by the lattice Kubo routine with the chosen Berry-curvature and Brillouin-zone orientation conventions. It should not be inferred from the sign of κ alone without fixing those conventions.

3 Mean-field branches

The self-consistent equations are solved at $\kappa = 0$. At this stage the Hamiltonian contains the off-diagonal kernel $\Phi(\mathbf{k})$ and the mean-field mass

$$m_Q(\mathbf{k}) = 2J\sum_\alpha Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \tag{134}$$

but not the diagnostic mass m_κ. This separation is useful because the signs of Q_α are selected self-consistently, whereas the signs in m_κ are fixed by the real-space loop orientation of Sec. 2.2.

$\mathbb{Z}_2$ at $\kappa = 0$. At $\kappa = 0$, changing the sign of all three Q parameters sends $m_Q(\mathbf{k}) \to -m_Q(\mathbf{k})$ while leaving $\Phi(\mathbf{k})$ unchanged. Hence

$$\omega_0(\mathbf{k}) = \sqrt{m_Q(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2} \tag{135}$$

is unchanged. Equivalently,

$$\mathcal{H}(\mathbf{k}; -Q, \kappa{=}0) = \sigma_x\, \mathcal{H}(\mathbf{k}; Q, \kappa{=}0)^*\, \sigma_x. \tag{136}$$

This is a relation between two mean-field kernels; it is not an additional microscopic lattice symmetry.

The covariance of the SCF map follows directly from the lower-band expectation values derived in Sec. 2.3. The A–B expectation values, which determine P_α and S_a, contain $u_A^* u_B = i\Phi^*/(2\omega_0)$ and are even under $Q \to -Q$. The same-sublattice expectation values, which determine Q_α, contain $|u_A|^2 - |u_B|^2 = m_Q/\omega_0$ and are odd under $Q \to -Q$. Therefore

$$\boxed{P_\alpha^{\rm out}(-Q) = P_\alpha^{\rm out}(Q), \qquad Q_\alpha^{\rm out}(-Q) = -Q_\alpha^{\rm out}(Q), \qquad S_a^{\rm out}(-Q) = S_a^{\rm out}(Q).} \tag{137}$$

Thus if (P^*, Q^*, S^*) is a fixed point, then $(P^*, -Q^*, S^*)$ is also a fixed point. Their $\kappa = 0$ energies are equal: the constant term is quadratic in Q, and the band energy depends on Q only through m_Q^2.

Finite κ. After the SCF fixed point has been obtained, the diagnostic hopping adds

$$m_\kappa(\mathbf{k}) = 2\kappa M_0(\mathbf{k}), \qquad M_0(\mathbf{k}) = \sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin(\mathbf{k}\cdot(\mathbf{R}_1{-}\mathbf{R}_2)) - \sin(\mathbf{k}\cdot\mathbf{R}_2). \tag{138}$$

The two $\mathbb{Z}_2$ partners have masses $m_Q(\mathbf{k}) + 2\kappa M_0(\mathbf{k})$ and $-m_Q(\mathbf{k}) + 2\kappa M_0(\mathbf{k})$. Labelling them as the $+Q^*$ and $-Q^*$ branches gives

$$\omega^2(\mathbf{k}; \pm Q^*, \kappa) = \omega_0^2(\mathbf{k}) \pm 4\kappa m_{Q^*}(\mathbf{k}) M_0(\mathbf{k}) + 4\kappa^2 M_0(\mathbf{k})^2, \qquad \omega_0^2 = m_{Q^*}^2 + |\Phi|^2. \tag{139}$$

Expanding the square root,

$$\omega(\mathbf{k}; \pm Q^*, \kappa) = \omega_0(\mathbf{k}) \pm \frac{2\kappa m_{Q^*}(\mathbf{k}) M_0(\mathbf{k})}{\omega_0(\mathbf{k})} + O(\kappa^2). \tag{140}$$

The constant part of the ground-state energy is the same on the two branches, so the leading splitting comes only from the band-energy sum:

$$\boxed{\Delta E_{\rm GS}(\kappa) \equiv E_{\rm GS}(+Q^*, \kappa) - E_{\rm GS}(-Q^*, \kappa) = -\frac{4\kappa}{N_k}\sum_{\mathbf{k}} \frac{m_{Q^*}(\mathbf{k}) M_0(\mathbf{k})}{\omega_0(\mathbf{k})} + O(\kappa^2).} \tag{141}$$

This formula gives a direct numerical check of the branch selection: for a small positive κ, the sign of the Brillouin-zone average fixes which partner has lower post-SCF energy.

Branch choice. Since κ is a post-SCF diagnostic hopping, it does not feed back into the self-consistent equations. Both (P^*, Q^*, S^*) and $(P^*, -Q^*, S^*)$ remain valid $\kappa = 0$ SCF solutions; the diagnostic spectrum is evaluated on the partner with the lower post-SCF energy at the chosen κ. Equivalently, for $\kappa > 0$ we use the branch continuously connected to the $\kappa \to 0^+$ ground state. All reported gaps, Chern numbers, and ribbon spectra use this selected representative.

4 Channel-level derivation

Section 2 gives the final mean-field kernel. This section keeps the channel-level derivation because it is the safest place to check which contraction produces each term and which constant accompanies it. The purpose is not to introduce new conventions, but to recover the already stated pieces of $\Phi(\mathbf{k})$, $m_J(\mathbf{k})$, and E_{const} from the direct, exchange, and cross channels separately.

4.1 NN hopping renormalization

The direct channel of the Fierz decomposition has the operator structure $(jk)(\ell m) = (AB)(AB)$ with coefficient $+J$, a product of two NN A–B bilinears. Mean-field decoupling shifts the bare NN coupling $K_1 \to K_1 + 2JP_\alpha$ on each bond type α.

Decoupling. The direct contraction reads

$$(ic_j c_k)(ic_\ell c_m) \xrightarrow{\text{Ch1 MF}} \langle ic_j c_k\rangle\,(ic_\ell c_m) + (ic_j c_k)\,\langle ic_\ell c_m\rangle - \langle ic_j c_k\rangle\langle ic_\ell c_m\rangle, \tag{142}$$

with (j,k) and (ℓ,m) the two NN A–B bilinears in $H_J^{(\alpha)}$ from (20)–(22). Both bilinears sit at the same cell displacement $\mathbf{\Delta}_\alpha$, so their expectation values are equal by translational invariance:

$$P_\alpha \equiv \big\langle ic_A(\mathbf{r})\, c_B(\mathbf{r}{+}\mathbf{\Delta}_\alpha)\big\rangle, \qquad \alpha = x, y, z, \tag{143}$$

real by Hermiticity of $ic_A c_B$ on distinct sites. The two operator terms in (142) carry the same coefficient JP_α and become identical lattice sums after a shift $\mathbf{r} \to \mathbf{r} - \mathbf{p}_\alpha$ in the second:

$$H_{\text{MF}}^{(\text{Ch1},\alpha)} = 2JP_\alpha \sum_{\mathbf{r}} ic_A(\mathbf{r})\, c_B(\mathbf{r}{+}\mathbf{\Delta}_\alpha) \;-\; NJP_\alpha^2, \tag{144}$$

where N is the number of unit cells.

Kernel. Summing (144) over α gives three NN A–B bilinears, each at its own cell displacement:

Bond	Cell disp $\mathbf{\Delta}_\alpha$	Bilinear (with coefficient)
x	$\mathbf{0}$	$2JP_x \sum_{\mathbf{r}} ic_A(\mathbf{r})\, c_B(\mathbf{r})$
y	$-\mathbf{R}_1$	$2JP_y \sum_{\mathbf{r}} ic_A(\mathbf{r})\, c_B(\mathbf{r}{-}\mathbf{R}_1)$
z	$-\mathbf{R}_2$	$2JP_z \sum_{\mathbf{r}} ic_A(\mathbf{r})\, c_B(\mathbf{r}{-}\mathbf{R}_2)$

These are the three NN A–B hoppings, each with K_1 shifted to $K_1 + 2JP_\alpha$. By the Bloch phase rule (an A–B bilinear at cell displacement $\boldsymbol{\Delta}$ contributes $+(\text{coefficient})\, e^{+i\mathbf{k}\cdot\boldsymbol{\Delta}}$ to $\Phi(\mathbf{k})$):

$$\boxed{\Phi_{\mathrm{Ch1}}(\mathbf{k}) = (K_1 + 2JP_x) + (K_1 + 2JP_y)\, e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K_1 + 2JP_z)\, e^{-i\mathbf{k}\cdot\mathbf{R}_2}.} \tag{145}$$

At $J = 0$ this is the bare Kitaev kernel $\Phi_0(\mathbf{k})$.

Constant. Summing $-NJP_\alpha^2$ over the three bond types:

$$E_{\mathrm{const}}^{(\mathrm{Ch1})} = -NJ\big(P_x^2 + P_y^2 + P_z^2\big). \tag{146}$$

4.2 Connection to the C_3 reduction

No independent C_3 convention is introduced in the channel checks. Applying

$$P_x = P_y = P_z \equiv P, \qquad (Q_z, Q_x, Q_y) = (Q, -Q, -Q), \qquad S_1 = S_2 = S_3 \equiv U \tag{147}$$

to the channel-level formulas reproduces the compact result of Sec. 2.4. In particular, the direct channel gives $\widetilde{K} = K_1 + 2JP$, the exchange channel gives the mass proportional to JQ, and the cross channel gives pairwise sums $S_i + S_j = 2U$. Since $m_\kappa(\mathbf{k})$ has the same sine form factor as the C_3-symmetric mean-field mass, the diagonal mass in this sector depends on the single combination $JQ + \kappa$.

At $J = 0$, the physical kernel is independent of the mean-field parameters because every mean-field correction is multiplied by J. The remaining mass is the post-SCF diagnostic Haldane mass in Eq. (133). The sign of the corresponding Chern number is determined with the Kubo and BZ-orientation conventions stated in Sec. 2.4.

4.3 Sublattice mass

The exchange channel of the Fierz decomposition has the operator structure $(j\ell)(km) = (AA)(BB)$ with coefficient $-J$, a product of one A–A bilinear and one B–B bilinear. Its role here is to derive the self-consistent mean-field mass $m_J(\mathbf{k})$ and its constant term. The post-SCF diagnostic mass $m_\kappa(\mathbf{k})$ was fixed in Sec. 2.2 and is not part of this decoupling.

A/B parameters. The exchange contraction of $H_J^{(\alpha)}$ in (20)–(22) reads

$$-J(ic_jc_\ell)(ic_kc_m) \xrightarrow{\text{Ch2 MF}} -J\big[\langle ic_jc_\ell\rangle(ic_kc_m) + (ic_jc_\ell)\langle ic_kc_m\rangle - \langle ic_jc_\ell\rangle\langle ic_kc_m\rangle\big]. \tag{148}$$

The pair (j, ℓ) connects the two A sites of the bond pair, at cell displacement $\mathbf{p}_\alpha$. The pair (k, m) connects the two B sites, also at cell displacement $\mathbf{p}_\alpha$. The two expectation values are not equal in general, so we introduce two real parameters per direction:

$$Q_{A,\alpha} \equiv \big\langle ic_A(\mathbf{r})\, c_A(\mathbf{r}+\mathbf{p}_\alpha)\big\rangle, \qquad Q_{B,\alpha} \equiv \big\langle ic_B(\mathbf{r})\, c_B(\mathbf{r}+\mathbf{p}_\alpha)\big\rangle. \tag{149}$$

Kernel. Equation (148) produces, for each α, an A–A bilinear with coefficient $-JQ_{B,\alpha}$ along $\mathbf{p}_\alpha$ and a B–B bilinear with coefficient $-JQ_{A,\alpha}$ along the same direction. Fourier transforming the diagonal bilinears (a real-space coefficient λ in front of $\sum_{\mathbf{r}} ic_X(\mathbf{r})\, c_X(\mathbf{r}+\boldsymbol{\Delta})$ contributes $-2\lambda \sin(\mathbf{k}\cdot\boldsymbol{\Delta})$ to $[\mathcal{H}]_{XX}$, identical for both $X = A$ and $X = B$ since each diagonal bilinear is itself Hermitian and the forward/backward pieces combine into the imaginary part):

$$[H_{\mathrm{MF}}^{(\mathrm{Ch2})}]_{AA} = +2J\sum_{\alpha} Q_{B,\alpha} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \qquad [H_{\mathrm{MF}}^{(\mathrm{Ch2})}]_{BB} = +2J\sum_{\alpha} Q_{A,\alpha} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{150}$$

The constant from the third term of (148), summed over α and unit cells, is $+JN\sum_\alpha Q_{A,\alpha}Q_{B,\alpha}$.

Stationarity. Diagonalising the full 2×2 two-band Majorana kernel gives the lower-band energy

$$E_-(\mathbf{k}) = J\sum_{\alpha}(Q_{A,\alpha}+Q_{B,\alpha}) \sin(\mathbf{k}\cdot\mathbf{p}_\alpha) - \Omega(\mathbf{k}), \qquad \Omega(\mathbf{k}) \equiv \sqrt{m_-(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2}, \tag{151}$$

where

$$m_-(\mathbf{k}) = J\sum_{\alpha}(Q_{A,\alpha}-Q_{B,\alpha}) \sin(\mathbf{k}\cdot\mathbf{p}_\alpha). \tag{152}$$

The total ground-state energy is

$$E_{\mathrm{GS}} = JN\sum_{\alpha} Q_{A,\alpha}Q_{B,\alpha} + \sum_{\mathbf{k}} E_-(\mathbf{k}). \tag{153}$$

The first term of E_- vanishes after summing over the BZ since $\sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha) = 0$, leaving only the constant and $-\sum_{\mathbf{k}}\Omega$. Since Ω depends on $Q_{A,\alpha}$ and $Q_{B,\alpha}$ only through their difference,

$$\frac{\partial\Omega}{\partial Q_{B,\alpha}} = -\frac{\partial\Omega}{\partial Q_{A,\alpha}}. \tag{154}$$

Stationarity of E_{GS} in $Q_{A,\alpha}$ and $Q_{B,\alpha}$ gives

$$JNQ_{B,\alpha} - \frac{\partial\sum_{\mathbf{k}}\Omega}{\partial Q_{A,\alpha}} = 0, \qquad JNQ_{A,\alpha} + \frac{\partial\sum_{\mathbf{k}}\Omega}{\partial Q_{A,\alpha}} = 0. \tag{155}$$

Adding the two equations cancels the Ω derivative:

$$\boxed{JN(Q_{A,\alpha}+Q_{B,\alpha}) = 0 \implies Q_{B,\alpha} = -Q_{A,\alpha} \equiv -Q_\alpha, \qquad \alpha = x,y,z.} \tag{156}$$

The relation holds for each NNN direction independently and reduces the six $Q_{A/B,\alpha}$ to three real Q_α.

Mass. Substituting (156) into (150) makes the two diagonal entries opposite,

$$[H_{\mathrm{MF}}^{(\mathrm{Ch2})}]_{AA} = -m_J(\mathbf{k}), \qquad [H_{\mathrm{MF}}^{(\mathrm{Ch2})}]_{BB} = +m_J(\mathbf{k}), \qquad m_J(\mathbf{k}) = 2J\sum_{\alpha} Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \tag{157}$$

each pairing direction $\mathbf{p}_z = \mathbf{R}_1$, $\mathbf{p}_x = \mathbf{R}_1-\mathbf{R}_2$, $\mathbf{p}_y = \mathbf{R}_2$ entering with a common positive sign, so that the chirality of the Haldane mass is carried entirely by the signs of the mean-field parameters Q_α. The three pairing vectors do not themselves close into a loop, $\mathbf{p}_z + \mathbf{p}_x + \mathbf{p}_y = 2\mathbf{R}_1$, whereas the closed next-nearest-neighbour loop is

$$\mathbf{p}_x + \mathbf{p}_y - \mathbf{p}_z = 0. \tag{158}$$

A Haldane mass is odd under $\mathbf{k} \to -\mathbf{k}$ and must be built on this closed loop; in (157) the closure is realised by the parameter signs rather than by fixed signs in front of the sine terms.

Constant. Using $Q_{B,\alpha} = -Q_{A,\alpha} \equiv -Q_\alpha$ in the constant $+JN\sum_\alpha Q_{A,\alpha}Q_{B,\alpha}$ gives

$$E_{\text{const}}^{(\text{Ch2})} = -NJ\big(Q_x^2 + Q_y^2 + Q_z^2\big). \tag{159}$$

Together with Eq. (157), this is the complete exchange-channel contribution to the self-consistent kernel. Any additional κ mass is added only after the SCF solution is obtained.

4.4 Off-diagonal 3NN correction

The cross channel of the Fierz decomposition has the operator structure $(jm)(k\ell) = (AB)(BA)$ with coefficient $+J$, a product of one A–B bilinear and one B–A bilinear. Mean-field decoupling generates two 3NN bilinears for each bond type, and the three bond types overlap on common 3NN A–B bilinears. The result is a $\mathbf{k}$-dependent off-diagonal correction $\Delta\Phi^{(\text{Ch3})}(\mathbf{k})$ whose three coefficients are pairwise sums of three independent parameters S_1, S_2, S_3.

Operators. The mean-field reduction reads

$$+J(ic_jc_m)(ic_kc_\ell) \xrightarrow{\text{Ch3 MF}} +J\big[\langle ic_jc_m\rangle\,(ic_kc_\ell) + (ic_jc_m)\,\langle ic_kc_\ell\rangle - \langle ic_jc_m\rangle\langle ic_kc_\ell\rangle\big], \tag{160}$$

with j, k, ℓ, m from the site table of §1. The pair (j, m) connects the A site of the first bond to the B site of the second bond, an A–B 3NN bilinear; the pair (k, ℓ) connects the B site of the first bond to the A site of the second bond, a B–A 3NN bilinear.

The cell displacements of these two bilinears can be read off explicitly. Take $\alpha = x$. From (20), $j = A(\mathbf{r})$ and $m = B(\mathbf{r} + \mathbf{R}_1 - \mathbf{R}_2)$, so (j, m) is at

$$\mathbf{p}_x + \boldsymbol{\Delta}_x = (\mathbf{R}_1 - \mathbf{R}_2) + \mathbf{0} = \mathbf{R}_1 - \mathbf{R}_2. \tag{161}$$

Similarly $k = B(\mathbf{r})$ and $\ell = A(\mathbf{r} + \mathbf{R}_1 - \mathbf{R}_2)$, so (k, ℓ) is a B–A bilinear at $\mathbf{R}_1 - \mathbf{R}_2 = \mathbf{p}_x - \boldsymbol{\Delta}_x$. To express its expectation value in the A–B reference orientation, use Majorana anticommutation:

$$\begin{aligned}\big\langle ic_kc_\ell\big\rangle &= \big\langle ic_B(\mathbf{r})c_A(\mathbf{r} + \mathbf{R}_1 - \mathbf{R}_2)\big\rangle \\ &= -\big\langle ic_A(\mathbf{r})c_B(\mathbf{r} - \mathbf{R}_1 + \mathbf{R}_2)\big\rangle = -S_2.\end{aligned} \tag{162}$$

Thus the x-bond row gives $\langle ic_jc_m\rangle = S_1$ and $\langle ic_kc_\ell\rangle = -S_2$. The same pattern holds for y and z: (j, m) sits at $\mathbf{p}_\alpha + \boldsymbol{\Delta}_\alpha$ and (k, ℓ) at $\mathbf{p}_\alpha - \boldsymbol{\Delta}_\alpha$, but the B–A expectation value must always be converted back to the A–B reference orientation.

Parameters. The three independent A–B 3NN expectation values are

$$S_1 \equiv \big\langle ic_A(\mathbf{r})\,c_B(\mathbf{r}{+}\mathbf{R}_1{-}\mathbf{R}_2)\big\rangle, \quad S_2 \equiv \big\langle ic_A(\mathbf{r})\,c_B(\mathbf{r}{+}\mathbf{R}_2{-}\mathbf{R}_1)\big\rangle, \quad S_3 \equiv \big\langle ic_A(\mathbf{r})\,c_B(\mathbf{r}{-}\mathbf{R}_1{-}\mathbf{R}_2)\big\rangle. \tag{163}$$

B–A expectation values also appear in (160), but they reduce to the S_a via Majorana anticommutation and translational invariance:

$$\boxed{\big\langle ic_B(\mathbf{r})\,c_A(\mathbf{r}{+}\boldsymbol{\Delta})\big\rangle = -\big\langle ic_A(\mathbf{r})\,c_B(\mathbf{r}{-}\boldsymbol{\Delta})\big\rangle} \qquad \text{(distinct sites)}. \tag{164}$$

Proof. Use anticommutation first, then translate the dummy cell label:

$$\begin{aligned}\big\langle ic_B(\mathbf{r})c_A(\mathbf{r}+\boldsymbol{\Delta})\big\rangle &= -\big\langle ic_A(\mathbf{r}+\boldsymbol{\Delta})c_B(\mathbf{r})\big\rangle \\ &= -\big\langle ic_A(\mathbf{r})c_B(\mathbf{r}-\boldsymbol{\Delta})\big\rangle.\end{aligned}$$

This proves the stated relation. □

Contributions. Reading off the cell displacements $\mathbf{p}_\alpha \pm \boldsymbol{\Delta}_\alpha$ for each bond and applying (163) and (164):

α	(j,m) disp $\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha$	$\langle ic_jc_m\rangle$	(k,ℓ) disp $\mathbf{p}_\alpha-\boldsymbol{\Delta}_\alpha$	$\langle ic_kc_\ell\rangle$
x	$\mathbf{R}_1-\mathbf{R}_2$	S_1	$\mathbf{R}_1-\mathbf{R}_2$	$-S_2$
y	$\mathbf{R}_2-\mathbf{R}_1$	S_2	$\mathbf{R}_1+\mathbf{R}_2$	$-S_3$
z	$\mathbf{R}_1-\mathbf{R}_2$	S_1	$\mathbf{R}_1+\mathbf{R}_2$	$-S_3$

The $-S_a$ entries follow from (164): for x with $\boldsymbol{\Delta} = \mathbf{R}_1 - \mathbf{R}_2$, and for y and z with $\boldsymbol{\Delta} = \mathbf{R}_1 + \mathbf{R}_2$.

The two operator terms in (160) contribute to $\Phi(\mathbf{k})$ as follows. Term 1 is a B–A bilinear at displacement $\mathbf{p}_\alpha - \boldsymbol{\Delta}_\alpha$ with coefficient $J\langle ic_jc_m\rangle$. The Bloch phase rule for B–A bilinears (an extra minus from anticommutation) gives

$$-J\langle ic_jc_m\rangle\, e^{-i\mathbf{k}\cdot(\mathbf{p}_\alpha-\boldsymbol{\Delta}_\alpha)}. \tag{165}$$

Term 2 is an A–B bilinear at displacement $\mathbf{p}_\alpha + \boldsymbol{\Delta}_\alpha$ with coefficient $J\langle ic_kc_\ell\rangle$, contributing

$$+J\langle ic_kc_\ell\rangle\, e^{+i\mathbf{k}\cdot(\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha)}. \tag{166}$$

The six contributions are

α	Term 1 (B–A)	Term 2 (A–B)
x	$-JS_1\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$	$-JS_2\, e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$
y	$-JS_2\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}$	$-JS_3\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$
z	$-JS_1\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}$	$-JS_3\, e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$

The bond-pair constants from the third term of (160) are $+JNS_1S_2$ for x, $+JNS_2S_3$ for y, and $+JNS_1S_3$ for z.

Total. Three distinct phase factors appear: $e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$, $e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$, $e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}$. Collecting their coefficients:

$$\boxed{\Delta\Phi^{(\mathrm{Ch3})}(\mathbf{k}) = -J\Big[(S_2{+}S_3)\, e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + (S_1{+}S_3)\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + (S_1{+}S_2)\, e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}\Big].} \tag{167}$$

The pairwise-sum coefficients reflect overlap: an A–B bilinear at a given cell displacement is generated by two of the three bond pairs. For example, the bilinear at $\mathbf{R}_1 - \mathbf{R}_2$ appears as $-JS_2$ (from x) and $-JS_3$ (from z), giving $-J(S_2 + S_3)$ on $e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)}$. The Channel-3 constant is

$$E_{\mathrm{const}}^{(\mathrm{Ch3})} = +JN\big(S_1S_2 + S_1S_3 + S_2S_3\big). \tag{168}$$

C_3 reduction. With $S_1 = S_2 = S_3 \equiv U$, each pairwise sum collapses to $2U$:

$$\Delta\Phi^{(\mathrm{Ch3})}(\mathbf{k})\big|_{C_3} = -2JU\Big[e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}\Big], \tag{169}$$

with constant $E^{(\mathrm{Ch3})}_{\mathrm{const}}|_{C_3} = +3JNU^2$.

5 Zigzag ribbon

Edge spectra are probed by passing to a finite-width ribbon. We adopt a zigzag termination, periodic along $\mathbf{R}_1$ and open along $\mathbf{R}_2$. The construction below produces a Hermitian Majorana matrix of size $2L \times 2L$,

$$H_{\mathrm{rib}}(k_x) \in \mathbb{C}^{2L\times 2L}, \tag{170}$$

for each longitudinal momentum k_x along $\mathbf{R}_1$, with L the ribbon width measured in unit cells along $\mathbf{R}_2$.

5.1 Geometry

Each site is labelled by an integer pair (n_1, ℓ) together with a sublattice index $X \in \{A, B\}$, where $n_1 \in \mathbb{Z}$ runs along $\mathbf{R}_1$ and $\ell \in \{0, 1, \ldots, L-1\}$ labels the layer along $\mathbf{R}_2$. Site positions inherit (13):

$$\mathbf{r}_A(n_1, \ell) = n_1\mathbf{R}_1 + \ell\mathbf{R}_2, \qquad \mathbf{r}_B(n_1, \ell) = \mathbf{r}_A(n_1, \ell) + \mathbf{s}_x. \tag{171}$$

Translational invariance survives only along $\mathbf{R}_1$, so we Bloch transform the index n_1 and retain ℓ as a layer block index:

$$c_X(n_1, \ell) = \frac{1}{\sqrt{N}} \sum_{k_x} e^{+ik_x\, n_1\sqrt{3}}\, c_{X,\ell}(k_x), \qquad X \in \{A, B\}, \tag{172}$$

with N the number of unit cells along $\mathbf{R}_1$ and the longitudinal phase

$$k_x\, n_1\sqrt{3} = k_x\, (n_1\, |\mathbf{R}_1|), \tag{173}$$

since $|\mathbf{R}_1| = \sqrt{3}$.

Longitudinal phase from a Majorana bilinear. The ribbon phase should be read in the same way as the bulk phase: it is fixed by the cell displacement of the real-space Majorana bilinear. Let a term connect

$$X(n_1, \ell) \quad \text{to} \quad Y(n_1 + a, \ell + b),$$

where a is the shift along $\mathbf{R}_1$ and b is the shift in the open layer direction. The corresponding oriented Majorana bilinear is

$$H^{(a,b)}_{XY} = i\alpha \sum_{n_1,\ell} c_X(n_1, \ell) c_Y(n_1 + a, \ell + b), \tag{174}$$

where α is real. Insert the one-dimensional Bloch transform (172):

$$\begin{aligned} H^{(a,b)}_{XY} &= \frac{i\alpha}{N} \sum_{n_1,\ell} \sum_{k_x,q_x} e^{+ik_x n_1\sqrt{3}} e^{+iq_x(n_1+a)\sqrt{3}} c_{X,\ell}(k_x) c_{Y,\ell+b}(q_x) \\ &= i\alpha \sum_{\ell} \sum_{k_x} e^{-ik_x a\sqrt{3}} c_{X,\ell}(k_x) c_{Y,\ell+b}(-k_x). \end{aligned} \tag{175}$$

In the k_x-block kernel, it is more convenient to place the first operator at momentum $-k_x$ and the second at momentum $+k_x$. Relabelling $k_x \mapsto -k_x$ gives

$$H_{XY}^{(a,b)} = i\alpha \sum_{\ell} \sum_{k_x} e^{+ik_x a\sqrt{3}} c_{X,\ell}(-k_x) c_{Y,\ell+b}(k_x). \tag{176}$$

Thus the longitudinal Bloch factor carried by a cell displacement $a\mathbf{R}_1 + b\mathbf{R}_2$ is

$$\phi_a(k_x) = e^{+ik_x a\sqrt{3}}. \tag{177}$$

The transverse shift b does not produce a Bloch phase in an open ribbon; it only tells which layer block is connected. For the displacements that appear in the present mean-field Hamiltonian, only three longitudinal shifts occur:

$$\boxed{\phi_0 \equiv 1, \qquad \phi_+ \equiv e^{+ik_x\sqrt{3}}, \qquad \phi_- \equiv e^{-ik_x\sqrt{3}}.} \tag{178}$$

This is the ribbon version of the bulk rule $e^{+i\mathbf{k}\cdot\boldsymbol{\Delta}}$: if the ribbon is later closed in the $\mathbf{R}_2$ direction, the missing transverse phase is restored and $e^{+ik_x a\sqrt{3}}$ becomes $e^{+i\mathbf{k}\cdot(a\mathbf{R}_1+b\mathbf{R}_2)}$.

5.2 Matrix entries

Reading off the longitudinal shift a and layer shift b from each cell displacement $\boldsymbol{\Delta} = a\mathbf{R}_1 + b\mathbf{R}_2$ in §1.2, and using the coefficients in (68) and (69), gives the following table.

Term	$\boldsymbol{\Delta}$ (cell disp)	$(\Delta n_1, \Delta\ell)$	$\phi(k_x)$	coefficient α
P_x (A–B)	$\mathbf{0}$	$(0,0)$	ϕ_0	$K_1 + 2JP_x$
P_y (A–B)	$-\mathbf{R}_1$	$(-1,0)$	ϕ_-	$K_1 + 2JP_y$
P_z (A–B)	$-\mathbf{R}_2$	$(0,-1)$	ϕ_0	$K_1 + 2JP_z$
$c_1 = -J(S_2{+}S_3)$ (A–B)	$\mathbf{R}_1{-}\mathbf{R}_2$	$(+1,-1)$	ϕ_+	c_1
$c_2 = -J(S_1{+}S_2)$ (A–B)	$-\mathbf{R}_1{-}\mathbf{R}_2$	$(-1,-1)$	ϕ_-	c_2
$c_3 = -J(S_1{+}S_3)$ (A–B)	$\mathbf{R}_2{-}\mathbf{R}_1$	$(-1,+1)$	ϕ_-	c_3
Q_z (A–A)	$\mathbf{R}_1$	$(+1,0)$	ϕ_+	$JQ_z + \kappa$
Q_x (A–A)	$\mathbf{R}_1{-}\mathbf{R}_2$	$(+1,-1)$	ϕ_+	$JQ_x - \kappa$
Q_y (A–A)	$\mathbf{R}_2$	$(0,+1)$	ϕ_0	$JQ_y - \kappa$

Each row is inserted into $H_{\mathrm{rib}}(k_x)$ by applying (176) to the forward operator and then adding its reversed partner. For A–B and A–A forward bilinears the matrix entry receives $+i\alpha\phi_a(k_x)$. For B–B forward bilinears the coefficient has the opposite sign, because the mass channel is sublattice-staggered

in (58). Explicitly,

$$\boxed{\begin{array}{lll} \text{A–B forward:} & [H_{\rm rib}]_{A_\ell,B_{\ell+\Delta\ell}} \mathrel{+}= +i\alpha\,\phi(k_x), & [H_{\rm rib}]_{B_{\ell+\Delta\ell},A_\ell} \mathrel{+}= -i\alpha\,\phi^*(k_x), \\ \text{A–A forward:} & [H_{\rm rib}]_{A_\ell,A_{\ell+\Delta\ell}} \mathrel{+}= +i\alpha\,\phi(k_x), & [H_{\rm rib}]_{A_{\ell+\Delta\ell},A_\ell} \mathrel{+}= -i\alpha\,\phi^*(k_x), \\ \text{B–B forward:} & [H_{\rm rib}]_{B_\ell,B_{\ell+\Delta\ell}} \mathrel{+}= -i\alpha\,\phi(k_x), & [H_{\rm rib}]_{B_{\ell+\Delta\ell},B_\ell} \mathrel{+}= +i\alpha\,\phi^*(k_x). \end{array}} \tag{179}$$

Two rows show how the table should be read. The P_y row has $\mathbf{\Delta}_y = -\mathbf{R}_1$, hence $(\Delta n_1, \Delta\ell) = (-1, 0)$ and $\phi = \phi_- = e^{-ik_x\sqrt{3}}$. It inserts

$$[H_{\rm rib}]_{A_\ell,B_\ell} \mathrel{+}= i(K_1 + 2JP_y)e^{-ik_x\sqrt{3}}, \qquad [H_{\rm rib}]_{B_\ell,A_\ell} \mathrel{+}= -i(K_1 + 2JP_y)e^{+ik_x\sqrt{3}}. \tag{180}$$

As a diagonal example, the Q_z row has $\mathbf{p}_z = \mathbf{R}_1$, so $(\Delta n_1, \Delta\ell) = (+1, 0)$ and $\alpha = m_z = JQ_z + \kappa$. The A–A and B–B insertions are

$$\begin{aligned} [H_{\rm rib}]_{A_\ell,A_\ell} &\mathrel{+}= +im_z e^{+ik_x\sqrt{3}}, \quad [H_{\rm rib}]_{A_\ell,A_\ell} \mathrel{+}= -im_z e^{-ik_x\sqrt{3}}, \\ [H_{\rm rib}]_{B_\ell,B_\ell} &\mathrel{+}= -im_z e^{+ik_x\sqrt{3}}, \quad [H_{\rm rib}]_{B_\ell,B_\ell} \mathrel{+}= +im_z e^{-ik_x\sqrt{3}}. \end{aligned} \tag{181}$$

Combining the two A–A entries gives $-2m_z \sin(k_x\sqrt{3})$, and combining the two B–B entries gives $+2m_z \sin(k_x\sqrt{3})$, matching the diagonal kernel convention $[-m(\mathbf{k}), +m(\mathbf{k})]$ after the transverse direction is closed.

The mass rows (Q_α) generate both an A–A insertion (coefficient $JQ_\alpha + \nu_\alpha\kappa$, with $\nu = (+, -, -)$ on (z, x, y)) and a partner B–B insertion (same coefficient, overall sign flipped), together reproducing the sublattice-staggered mass of (69). Open boundary conditions enter through the layer index: any term with $\ell + \Delta\ell \notin \{0, \ldots, L-1\}$ is dropped from the matrix. The resulting $H_{\rm rib}(k_x)$ is block-tridiagonal in ℓ, with each block of size 2 indexed by sublattice within a layer.

5.3 Checks

Hermiticity holds by construction: in each row of (179) swap of matrix indices is accompanied by $\phi \leftrightarrow \phi^*$:

$$[H_{\rm rib}]_{ij} = [H_{\rm rib}]^*_{ji} \qquad \text{for all } i, j. \tag{182}$$

Bulk recovery is verified by closing the ribbon along $\mathbf{R}_2$: treat ℓ as a cyclic index of period L and Bloch transform with phase $e^{+ik_\ell\ell}$. The 1D phase $\phi(k_x) = e^{+ik_x\Delta n_1\sqrt{3}}$ then becomes $e^{+i(k_x\Delta n_1\sqrt{3} + k_\ell\Delta\ell)}$. Identifying

$$k_x\sqrt{3} = \mathbf{k}\cdot\mathbf{R}_1, \qquad k_\ell = \mathbf{k}\cdot\mathbf{R}_2, \qquad \text{equivalently} \qquad k_y = \frac{2k_\ell - k_x\sqrt{3}}{3}, \tag{183}$$

recovers $e^{+i\mathbf{k}\cdot\mathbf{\Delta}}$ for every cell displacement $\mathbf{\Delta} = \Delta n_1\mathbf{R}_1 + \Delta\ell\mathbf{R}_2$ in the table.

This can be seen directly at the level of the two bulk matrix elements. Let

$$a \equiv k_x\sqrt{3} = \mathbf{k}\cdot\mathbf{R}_1, \qquad b \equiv k_\ell = \mathbf{k}\cdot\mathbf{R}_2, \tag{184}$$

and define

$$\begin{aligned} &\widetilde{K}_\alpha \equiv K_1 + 2JP_\alpha, \qquad m_z \equiv JQ_z + \kappa, \quad m_x \equiv JQ_x - \kappa, \quad m_y \equiv JQ_y - \kappa, \\ &c_1 \equiv -J(S_2 + S_3), \qquad c_2 \equiv -J(S_1 + S_2), \qquad c_3 \equiv -J(S_1 + S_3). \end{aligned} \tag{185}$$

After the transverse Bloch transform, the off-diagonal entry of the closed ribbon is

$$\begin{aligned}\Phi_{\mathrm{rib}}(a,b) = \widetilde{K}_x + \widetilde{K}_y e^{-ia} + \widetilde{K}_z e^{-ib} \\ + c_1 e^{+i(a-b)} + c_2 e^{-i(a+b)} + c_3 e^{-i(a-b)}.\end{aligned} \tag{186}$$

Using $a = \mathbf{k}\cdot\mathbf{R}_1$ and $b = \mathbf{k}\cdot\mathbf{R}_2$, this is exactly (68). The diagonal entries become

$$\begin{aligned}[H_{\mathrm{rib}}]_{AA} &= -2\big[m_z \sin a + m_x \sin(a-b) + m_y \sin b\big], \\ [H_{\mathrm{rib}}]_{BB} &= +2\big[m_z \sin a + m_x \sin(a-b) + m_y \sin b\big].\end{aligned} \tag{187}$$

Therefore the closed-ribbon diagonal part is $\mathrm{diag}[-m(\mathbf{k}), +m(\mathbf{k})]$ with $m(\mathbf{k})$ given by (69). Hence the closed zigzag ribbon recovers $\mathcal{H}(\mathbf{k})$ exactly under the identification $(k_x, k_\ell) \leftrightarrow \mathbf{k}$. Numerical verification at machine precision is recorded in Section 7.

6 Armchair ribbon

The armchair ribbon is the second edge geometry used in the code. The geometric idea is the same as in the zigzag construction: keep translation symmetry in one direction, keep the transverse coordinate as a layer index, and insert each real-space Majorana bilinear into a finite matrix. The only new feature is that a convenient armchair repeat cell contains two primitive honeycomb cells. Therefore the armchair ribbon has a four-site layer basis, and the closed-ribbon check must be compared with a folded bulk spectrum rather than with a single two-band bulk matrix.

The shaded cell in the armchair edge drawing should be read in precisely this way. The horizontal black arrow is the armchair periodic direction, while the vertical black arrow is the transverse direction. The red, green, and blue arrows still denote the original x, y, z bond directions, but the rectangular armchair cell groups two primitive cells before imposing one-dimensional Bloch periodicity.

6.1 Geometry

Rotate the real-space frame by 90° counterclockwise, $(x, y) \mapsto (-y, x)$. In this rotated frame the armchair direction is chosen as the periodic direction. The supercell vectors are

$$\widetilde{\mathbf{R}}_x^{\mathrm{arm}} = (3, 0), \qquad \widetilde{\mathbf{R}}_y^{\mathrm{arm}} = (0, \sqrt{3}). \tag{188}$$

The ribbon is periodic along $\widetilde{\mathbf{R}}_x^{\mathrm{arm}}$ and open along $\widetilde{\mathbf{R}}_y^{\mathrm{arm}}$. A site is labelled by an armchair cell index $N \in \mathbb{Z}$, a transverse layer index $\ell = 0, \ldots, L-1$, and one of four sites inside the armchair supercell.

The implementation uses the four-site order

$$\boxed{(B_2, A_2, B_1, A_1)_\ell \longleftrightarrow (4\ell, 4\ell+1, 4\ell+2, 4\ell+3).} \tag{189}$$

The labels $1, 2$ distinguish the two primitive cells inside one armchair supercell; they are not new sublattices. The order in (189) is a code basis convention. It may differ from the visual order in the edge

drawing, but all hopping rows below are written in this same basis, so the matrix is unambiguous. A convenient set of coordinates inside one armchair supercell, measured in the rotated frame of (188) with the origin at B_2, is

basis site	matrix index in layer ℓ	position inside the supercell
B_2	4ℓ	$(0, 0)$
A_2	$4\ell + 1$	$(1/2, \sqrt{3}/2)$
B_1	$4\ell + 2$	$(3/2, \sqrt{3}/2)$
A_1	$4\ell + 3$	$(2, 0)$

These coordinates are not an additional convention; they only make explicit the four-site order used by the code. Translating a target site by $\widetilde{\mathbf{R}}_x^{\mathrm{arm}} = (3, 0)$ changes N by $+1$, and translating by $\widetilde{\mathbf{R}}_y^{\mathrm{arm}} = (0, \sqrt{3})$ changes the layer index by $+1$.

The one-dimensional Bloch transform is

$$c_X(N, \ell) = \frac{1}{\sqrt{N_c}} \sum_{k_x} e^{+ik_x(3N)} c_{X,\ell}(k_x), \qquad X \in \{B_2, A_2, B_1, A_1\}, \tag{190}$$

where N_c is the number of armchair supercells along the periodic direction. Thus the one-dimensional Brillouin zone is $k_x \in [-\pi/3, \pi/3)$.

Longitudinal phase from a Majorana bilinear. As in the zigzag ribbon, the Bloch phase is fixed by the displacement of the real-space Majorana bilinear. Consider a forward row connecting

$$X(N, \ell) \quad \text{to} \quad Y(N + \Delta N, \ell + \Delta\ell)$$

with real coefficient α:

$$H_{XY}^{(\Delta N, \Delta\ell)} = i\alpha \sum_{N,\ell} c_X(N, \ell) c_Y(N + \Delta N, \ell + \Delta\ell). \tag{191}$$

Inserting (190) gives

$$\begin{aligned} H_{XY}^{(\Delta N, \Delta\ell)} &= \frac{i\alpha}{N_c} \sum_{N,\ell} \sum_{k_x, q_x} e^{+ik_x(3N)} e^{+iq_x 3(N+\Delta N)} c_{X,\ell}(k_x) c_{Y,\ell+\Delta\ell}(q_x) \\ &= i\alpha \sum_{\ell, k_x} e^{+i3k_x \Delta N} c_{X,\ell}(-k_x) c_{Y,\ell+\Delta\ell}(k_x). \end{aligned} \tag{192}$$

Therefore a forward row with armchair-cell shift ΔN carries

$$\phi_{\Delta N}(k_x) = e^{+i3k_x \Delta N}, \qquad \phi_+ \equiv e^{+i3k_x}, \quad \phi_- \equiv e^{-i3k_x}. \tag{193}$$

The layer shift $\Delta\ell$ does not produce a Bloch phase in the open ribbon; it only tells which transverse layer block is connected. This is the armchair analogue of (177).

6.2 Matrix entries

Use the same coefficient definitions as in the zigzag bulk-recovery check:

$$\begin{aligned} &\widetilde{K}_\alpha = K_1 + 2JP_\alpha, \qquad m_z = JQ_z + \kappa, \quad m_x = JQ_x - \kappa, \quad m_y = JQ_y - \kappa, \\ &c_1 = -J(S_2 + S_3), \qquad c_2 = -J(S_1 + S_2), \qquad c_3 = -J(S_1 + S_3). \end{aligned} \tag{194}$$

Each row in the table below means the forward real-space bilinear (191). The pair $(\Delta N, \Delta\ell)$ is read in the armchair supercell coordinates: ΔN gives the Bloch phase $e^{+i3k_x\Delta N}$, while $\Delta\ell$ gives the target layer. The Hermitian reverse row is not listed separately; it is inserted by the rule following the table.

row	sector	forward bilinear	$\Delta\ell$	ΔN	coefficient α
1	NN	$A_1 \to B_1$	0	0	$\widetilde{K}_x$
2	NN	$A_2 \to B_1$	0	0	$\widetilde{K}_z$
3	NN	$A_2 \to B_2$	0	0	$\widetilde{K}_y$
4	NN	$A_1 \to B_2$	0	$+1$	$\widetilde{K}_z$
5	NN	$A_1 \to B_1$	-1	0	$\widetilde{K}_y$
6	NN	$A_2 \to B_2$	$+1$	0	$\widetilde{K}_x$
7	3NN	$A_1 \to B_2$	0	0	c_3
8	3NN	$A_1 \to B_2$	$+1$	$+1$	c_1
9	3NN	$A_1 \to B_2$	-1	$+1$	c_2
10	3NN	$A_2 \to B_1$	0	-1	c_3
11	3NN	$A_2 \to B_1$	$+1$	0	c_1
12	3NN	$A_2 \to B_1$	-1	0	c_2
13	NNN-A	$A_1 \to A_1$	$+1$	0	m_z
14	NNN-A	$A_2 \to A_2$	$+1$	0	m_z
15	NNN-A	$A_1 \to A_2$	0	$+1$	m_x
16	NNN-A	$A_2 \to A_1$	$+1$	0	m_x
17	NNN-A	$A_1 \to A_2$	0	0	m_y
18	NNN-A	$A_2 \to A_1$	$+1$	-1	m_y
19	NNN-B	$B_2 \to B_2$	$+1$	0	m_z
20	NNN-B	$B_1 \to B_1$	$+1$	0	m_z
21	NNN-B	$B_2 \to B_1$	0	0	m_x
22	NNN-B	$B_1 \to B_2$	$+1$	$+1$	m_x
23	NNN-B	$B_2 \to B_1$	0	-1	m_y
24	NNN-B	$B_1 \to B_2$	$+1$	0	m_y

The matrix insertion follows directly from (192). For A–B and A–A forward rows, insert $+i\alpha\phi_{\Delta N}$ in the forward direction and the Hermitian reverse $-i\alpha\phi^*_{\Delta N}$. For B–B rows the sign is reversed, because

the same-sublattice mass is sublattice-staggered: the bulk kernel has diagonal entries $[-m(\mathbf{k}), +m(\mathbf{k})]$. Thus

$$\begin{aligned} \text{A–B or A–A:} \quad [H_{\text{arm}}]_{X_\ell, Y_{\ell+\Delta\ell}} &\mathrel{+}= +i\alpha e^{+i3k_x\Delta N}, \\ [H_{\text{arm}}]_{Y_{\ell+\Delta\ell}, X_\ell} &\mathrel{+}= -i\alpha e^{-i3k_x\Delta N}, \\ \text{B–B:} \quad [H_{\text{arm}}]_{B_\ell, B'_{\ell+\Delta\ell}} &\mathrel{+}= -i\alpha e^{+i3k_x\Delta N}, \\ [H_{\text{arm}}]_{B'_{\ell+\Delta\ell}, B_\ell} &\mathrel{+}= +i\alpha e^{-i3k_x\Delta N}. \end{aligned} \tag{195}$$

For example, row 4 is the NN bilinear $A_1(N,\ell) \to B_2(N+1,\ell)$ with coefficient $\widetilde{K}_z$. Since $\Delta N = +1$, it gives

$$[H_{\text{arm}}]_{A_{1,\ell}, B_{2,\ell}} \mathrel{+}= i\widetilde{K}_z e^{+i3k_x}, \qquad [H_{\text{arm}}]_{B_{2,\ell}, A_{1,\ell}} \mathrel{+}= -i\widetilde{K}_z e^{-i3k_x}. \tag{196}$$

As a same-sublattice example, row 23 is a B–B bilinear $B_2(N,\ell) \to B_1(N-1,\ell)$ with coefficient m_y. The B–B sign rule gives

$$[H_{\text{arm}}]_{B_{2,\ell}, B_{1,\ell}} \mathrel{+}= -im_y e^{-i3k_x}, \qquad [H_{\text{arm}}]_{B_{1,\ell}, B_{2,\ell}} \mathrel{+}= +im_y e^{+i3k_x}. \tag{197}$$

Rows whose target layer lies outside $0 \le \ell \le L-1$ are dropped for the open ribbon.

6.3 Bulk recovery

Closing the armchair ribbon along $\widetilde{\mathbf{R}}_y^{\text{arm}}$ gives a fully periodic system with the rectangular four-site supercell (188). Because this cell contains two primitive honeycomb cells, the closed armchair Hamiltonian is a folded four-band representation of the same bulk problem. Therefore the correct check is spectral: one armchair momentum block must reproduce two primitive bulk momenta.

Let the transverse Bloch phase be

$$q_\ell = \frac{2\pi n}{L}, \qquad n = 0, 1, \ldots, L-1. \tag{198}$$

The corresponding armchair-frame momenta are

$$\mathbf{k}_0^{\text{arm}} = (k_x, q_\ell/\sqrt{3}), \qquad \mathbf{k}_1^{\text{arm}} = \mathbf{k}_0^{\text{arm}} + (2\pi/3, 0). \tag{199}$$

The shift $(2\pi/3, 0)$ is a reciprocal vector of the armchair supercell but not of the primitive honeycomb cell. It is therefore the second primitive momentum folded into the same armchair supercell momentum. Returning to the original, unrotated frame uses the inverse of the 90° rotation,

$$(k_x^{\text{orig}}, k_y^{\text{orig}}) = (k_y^{\text{arm}}, -k_x^{\text{arm}}). \tag{200}$$

The bulk-recovery statement is

$$\operatorname{Spec} H_{\text{arm}}^{\text{PBC}}(k_x, q_\ell) = \operatorname{Spec} \mathcal{H}(\mathbf{k}_0^{\text{orig}}) \cup \operatorname{Spec} \mathcal{H}(\mathbf{k}_1^{\text{orig}}), \tag{201}$$

where $\mathbf{k}_{0,1}^{\text{orig}}$ are obtained from (199) by (200). This folded recovery is the armchair counterpart of the direct two-band recovery in the zigzag ribbon. Numerical checks of (201) give agreement at machine precision; see §7.

7 Numerical checks

The construction has been checked numerically against analytic limits and internal consistency conditions:

- **Pure-Kitaev limit ($J = 0$).** All mean-field corrections are multiplied by J, so the physical kernel is independent of P, Q, U and reduces to $\Phi(\mathbf{k}) = \Phi_0(\mathbf{k})$ plus the κ-only mass of (133). Numerical Chern computation returns $C_\kappa = -\mathrm{sgn}(\kappa)$ with the present Kubo and Brillouin-zone orientation conventions.
- $\kappa = 0$ **backward compatibility.** Bulk and ribbon outputs are bit-identical to the pre-κ implementation.
- **Hermiticity.** $\mathcal{H}(\mathbf{k})$ and $H_{\mathrm{rib}}(k_x)$ are Hermitian to machine precision for all parameter values, including finite κ.
- **Ribbon $\to$ bulk under PBC.** Closing the zigzag ribbon along $\mathbf{R}_2$ and identifying $k_\ell = \mathbf{k}\cdot\mathbf{R}_2$ as in (183) reproduces the two-band bulk spectrum to $\sim 10^{-15}$. Closing the armchair ribbon reproduces the folded four-band spectrum in (201) to $\sim 10^{-15}$, for all κ, J, and L tested.
- **Bulk–edge correspondence.** At $J = 1, \kappa = 0.2$ the post-SCF diagnostic spectrum has $C_\kappa = +2$, and the ribbon spectrum exhibits two pairs of chiral edge modes, one pair at each boundary, with opposite edge polarisation.

8 Summary

The three-bond extension promotes the c^4 interaction H_J from a single z-bond pair to all three bond types, introducing nine mean-field parameters: P_α (Channel 1), Q_α (Channel 2), and S_a (Channel 3). Channel 1 shifts the bare NN coupling $K_1 \to K_1 + 2JP_\alpha$ on each bond; under C_3 this is an isotropic rescaling of the bare kernel and the Dirac points are unmoved. Channel 2 generates a sublattice-staggered mass

$$m_J(\mathbf{k}) = 2J\sum_\alpha Q_\alpha \sin(\mathbf{k}\cdot\mathbf{p}_\alpha),$$

of Haldane type, with $Q_{B,\alpha} = -Q_{A,\alpha}$ enforced independently for each NNN direction by stationarity. Channel 3 contributes a $\mathbf{k}$-dependent off-diagonal correction whose three coefficients are pairwise sums $S_i + S_j$, with the B–A bilinears reduced to the same three S_a via Majorana anticommutation.

A separate piece of the diagonal mass is included as a post-SCF diagnostic time-reversal-breaking chiral NNN Majorana hopping. Its amplitude κ is not mean-field-decoupled and its microscopic coefficient is not fixed in this note. With the present orientation convention it enters as

$$m(\mathbf{k}) = 2\sum_{\alpha=z,x,y} (JQ_\alpha + \nu_\alpha\kappa)\sin(\mathbf{k}\cdot\mathbf{p}_\alpha), \qquad (\nu_z, \nu_x, \nu_y) = (+1, -1, -1).$$

Only in the C_3-symmetric sector, where $(Q_z, Q_x, Q_y) = (Q, -Q, -Q)$, does this reduce to the single combination $JQ + \kappa$. At $J = 0$, all self-consistent corrections drop out of the physical kernel, leaving the bare Haldane–Kitaev Majorana mass; with the present Kubo convention this gives $C_\kappa = -\mathrm{sgn}(\kappa)$.

At $J = 1, \kappa = 0.2$ the post-SCF diagnostic spectrum has a $C_\kappa = +2$ descendant whose ribbon spectrum carries two pairs of chiral edge modes.

The zigzag ribbon construction (Section 5) implements the same nine bilinears as a $2L \times 2L$ block-tridiagonal matrix $H_{\mathrm{rib}}(k_x)$, with the longitudinal phase $\phi(k_x) = e^{+ik_x \Delta n_1 \sqrt{3}}$ depending only on the cell shift along $\mathbf{R}_1$ and the layer index ℓ entering as a block label. The full parameter set is

$$(P_\alpha, Q_\alpha, S_a, K_1, J, \kappa, L)$$

in general, and

$$(P, Q, U, K_1, J, \kappa, L)$$

under C_3 symmetry.

The armchair ribbon construction (Section 6) uses a four-site supercell ordered as (B_2, A_2, B_1, A_1). Its periodic closure therefore does not recover a single two-band bulk matrix directly; instead it recovers the folded union of two bulk spectra, as in (201). This spectral recovery has been checked to machine precision, independently of the zigzag recovery test.